

The Effect of Class Size on Student Achievement

Instrumental variables estimation: a replication of Angrist & Lavy (1999) and an extension on social heterogeneity

Olivier GILLE

Master's Thesis (M2)

Supervised by Yves Croissant

Academic year 2025–2026

Abstract

This thesis is set against the background of the halving of class sizes in first and second grade in French priority education networks, introduced in 2017. The policy aims to improve the achievement of pupils from disadvantaged backgrounds. The thesis offers a review of the empirical literature on class size effects, sets out the methodological framework of the instrumental variables estimator, and replicates the results of Angrist & Lavy (1999). An original extension then decomposes the effect by terciles of disadvantage and examines it on an outcome absent from the original paper: the failure rate.

Keywords: class size; student achievement; instrumental variables; Maimonides' rule; regression discontinuity; priority education.

Introduction

In 2017, France introduced the halving of first- and second-grade (CP and CE1) class sizes in priority education networks (REP and REP+), reducing enrolment from roughly 22 to 13 pupils. Its cost, on the order of 700 million euros a year, makes a rigorous empirical evaluation of its effectiveness indispensable. The policy has one feature that governs this entire study: it does not reduce class size everywhere, but only in the country's most disadvantaged schools. Its justification therefore rests on a proposition borrowed from the literature, according to which the effect of class size is all the stronger the more disadvantaged the pupil population.

The question has generated an abundant literature whose conclusions do not always converge. The STAR experiment in Tennessee (Krueger, 1999) finds effects on the order of 2 to 3% of a standard deviation per pupil removed, but such experiments are rare and of limited external validity, as the Californian replication by Jepsen and Rivkin (2009) shows. In their absence, instrumental variables studies, which exploit administrative rules to generate exogenous variation in class size, constitute the most credible approach: Angrist and Lavy (1999) in Israel, Piketty and Valdenaire (2006) in France and Fredriksson et al. (2013) in Sweden all find significant effects, on the order of 1 to 3% of a standard deviation per pupil. These results are not universal, however: Hoxby (2000) detects no effect, Angrist, Battistin and Vuri (2017) show that the effect measured in Italy vanishes once a test-manipulation bias is neutralised, and Angrist et al. (2019) fail, twenty years on, to recover the effects they had themselves established in 1999.

This thesis first offers a review of that literature, updating the synthesis of Bouguen, Grenet and Gurgand (2017) by bringing all estimates onto a common scale: the gain associated with one fewer pupil, expressed as a percentage of a standard deviation. It then sets out the methodological framework of the instrumental variables estimator, before replicating Angrist and Lavy (1999) in Python and testing the robustness of the main estimate on a sample restricted to the neighbourhood of the thresholds.

The extension that follows is the thesis's own contribution, and it bears precisely on the proposition underpinning the French targeting. Angrist and Lavy test social heterogeneity through a single interaction term, which constrains the effect to vary linearly with the share of disadvantaged pupils; yet it is by extrapolating that slope that one would predict the gain expected in the most disadvantaged schools. Two independent approaches relax this assumption: a quadratic specification that tests curvature directly, and a decomposition by terciles that imposes no functional form and extends the approach of Piketty and Valdenaire (2006) with the formal test they lacked. Both approaches are then applied to an outcome absent from the original paper, the failure rate: Angrist and Lavy's estimation bears on the class *mean* score, whereas a policy targeted at disadvantaged backgrounds aims less at raising the mean than at reducing the share of struggling pupils. This distinction will prove decisive, and will lead to a new reading of the French evaluations of the class-size reduction policy.

Part I. Review of the empirical literature on class size effects

1 The main studies in the literature

1.1 Experimental studies

1.1.1 STAR: Krueger (1999) and Chetty et al. (2011)

The question of the effect of class size on student achievement is an old one. The STAR (Student/Teacher Achievement Ratio) experiment, conducted in Tennessee between 1985 and 1989, remains today the literature's principal reference.

Roughly 12,000 pupils from kindergarten to third grade were randomly assigned to three class types: small classes (13–17 pupils), large classes (22–26 pupils), and large classes with a teaching aide. Teachers were also randomly assigned. This protocol rules out selection bias by construction: any difference in outcomes between groups can be causally attributed to class size.

The results were analysed in two stages. Finn and Achilles (1990) provide a first evaluation and establish that pupils in small classes obtain better results in reading and mathematics, with an *effect size* of roughly 0.2 to 0.3 standard deviations. Their analysis nonetheless suffers from important methodological limitations: they do not correct standard errors for intra-class correlation and do not formally address the problem of non-random transitions between class types.

Krueger (1999) revisits the same data with a more rigorous methodology. The problem he seeks to solve is the following: about 10% of pupils changed class type during the experiment for non-random reasons (behavioural problems, parental requests), which biases the direct comparison between groups. To correct for this, rather than comparing pupils by the class they actually attended, he exploits the initial random assignment to reconstruct a purely exogenous source of class-size variation. In a first step, he estimates what class size each pupil should have had according to their initial assignment; in a second step, he uses this predicted size, rather than the actual size, to estimate the effect on scores. This procedure, known as two-stage least squares (2SLS, described in detail in Part II), neutralises the bias introduced by non-random transitions. His results indicate that a reduction of ten pupils is associated with a gain of 7 to 9 percentiles depending on the grade level.

Chetty et al. (2011) complement these studies by linking the STAR data to US tax records, which allows them to follow 95% of participants into adulthood and to observe outcomes hitherto inaccessible: college attendance, retirement savings, home ownership, neighbourhood quality. On class size, they find a significant effect on the probability of attending college at age 20 (+1.8 percentage points). Their main contribution concerns a measure of class quality, defined as the end-of-year mean scores of a pupil's classmates relative to other pupils in the same school, which simultaneously captures the teacher effect, peer effects and any shock common to the class. They document a paradox: the effects of this class quality on scores disappear by the fourth grade, yet they reappear as significant effects on adult earnings (+\$50.6

per year per percentile of quality, $p < 0.01$). Thus, although in this study teacher quality and peer composition have no effect on test scores, that effect is recovered on other outcomes. Class size is therefore only one of the channels that can affect pupils, and any identification strategy that fails to control for joint changes in teacher quality and peer composition risks attributing to class size alone effects that are in reality foreign to it.

1.1.2 The Californian reform: Jepsen & Rivkin (2009)

Jepsen and Rivkin (2009) illustrate this warning empirically by studying the Californian class-size reduction (CSR) programme, which cut class sizes from kindergarten to third grade from about 30 to 20 pupils starting in the summer of 1996, creating 25,000 new teaching posts in two years. The authors begin with difference-in-differences models (comparing schools that adopted CSR early and late) in order to facilitate comparison with earlier evaluations of the programme, but their preferred specification is a model with school-by-grade and school-by-year fixed effects, which they favour because it does not rest on the strong assumptions underlying difference-in-differences approaches. This model allows them to estimate simultaneously, within the same regression, the direct effect of class size and the indirect effect of the deterioration in teacher quality that accompanied the reform.

The model estimates two effects separately, each controlling for the other: on the one hand, the effect of class size, all else equal, on the order of 0.056 to 0.096 standard deviations in mathematics and 0.035 to 0.063 in reading (grades 2–4) for a reduction of ten pupils; on the other hand, the effect of teacher inexperience. In reality, however, CSR produced both changes at once: by reducing class sizes, it also forced the massive recruitment of novice teachers. By combining the two coefficients with the changes that actually occurred in California, the authors therefore compute the policy’s real net effect: 0.116 standard deviations in mathematics and 0.074 in reading in the short run (1995–1997), and 0.167 and 0.099 in the longer run (1995–2001), once the initial shock of novice teachers had been absorbed. This net effect remains positive but is markedly smaller than what the Tennessee results predicted, where teacher quality was not itself affected by the reform.

This gap is moreover unevenly distributed: high-poverty and high-minority schools, which experienced the largest increases in the share of inexperienced teachers, enjoyed a smaller net gain in the short run (about 20% less than low-poverty schools), a gap that nonetheless narrows substantially over time (to about 4–5% a few years later).

The authors acknowledge, however, that experience and certification capture only a fraction of real teacher quality, with unobservable quality (motivation, intrinsic pedagogical skill, fit with the post) remaining in the error term.

1.2 Instrumental variables studies

1.2.1 Angrist & Lavy (1999)

In the absence of a randomised experiment, researchers turn to quasi-experimental identification strategies. The most influential contribution is that of Angrist & Lavy (1999), who exploit Maimonides’ rule as an instrumental variable. This rule generates sharp discontinuities at

thresholds of 40, 80 and 120 pupils per cohort: a cohort of 40 pupils yields a single class of 40, whereas a cohort of 41 pupils yields two classes of about 20. These discontinuities are purely mechanical and independent of pupil characteristics.

We discuss the results of this study at length in our replication section.

1.2.2 Piketty & Valdenaire (2006)

Using French data, Piketty and Valdenaire (2006) propose an adaptation of Angrist & Lavy’s method. They use the 1997 primary panel (about 9,600 pupils entering first grade in autumn 1997, whose non-repeaters entered lower secondary school in 2002) and the 1995 secondary panel (about 17,800 pupils entering lower secondary school in autumn 1995, followed through to the baccalauréat in 2003), exploiting the discontinuities associated with implicit class-opening thresholds as instrumental variables. In France there is no uniform national threshold comparable to Maimonides’ rule, but the authors document very strong implicit threshold effects in the data: in primary school, the implicit threshold is around 30 pupils in second grade, and crossing it produces an abrupt drop in class size from 30 to about 15–16.

The results for primary school are substantial. A reduction of one pupil in second grade leads to an average increase of about 0.3 to 0.4 points in the mean mathematics score at the start of third grade, or roughly 2.5 to 3% of a standard deviation. For disadvantaged pupils, the effect rises to at least 0.7 points. Effects in lower and upper secondary school exist but are weaker, around 0.4% of a standard deviation, which the authors interpret as an argument for concentrating targeted policies on the youngest grades.

The way Piketty and Valdenaire establish this differentiated result deserves to be spelled out, as it will be central to our extension (§2). To show that the effect is stronger among disadvantaged pupils, the authors proceed by splitting the sample: they re-estimate their instrumental model separately on each social subgroup, then juxtapose the coefficients obtained. The 0.7-point gain mentioned above corresponds precisely to this split: in primary school, removing one pupil from the second-grade class raises the mathematics score by 0.715 points for pupils whose parents’ occupational category is disadvantaged, against only 0.274 points for children of advantaged occupational categories; likewise, the gain is 0.575 points for pupils whose mother does not hold the baccalauréat, against 0.261 points for those whose mother is a graduate. They conclude that the effect is markedly more pronounced among disadvantaged pupils. This conclusion rests on the direct comparison of coefficients across subgroups, without any formal test of their equality.

1.2.3 Fredriksson, Öckert and Oosterbeek (2013)

Fredriksson, Öckert and Oosterbeek (2013) also find long-run effects in Sweden: beyond cognitive gains (logical and verbal skills), estimated at 3.30% of a standard deviation per pupil removed between fourth and sixth grade, pupils educated in small classes obtain higher earnings in adulthood, with an increase of about 0.63% per pupil removed.

1.3 Other studies included in the review by Bouguen, Grenet, and Gurgand (2017)

Bressoux, Kramarz and Prost (2009) exploit administrative forecasting errors that led, in 1991, some novice teachers to take up their posts without training. Their identification rests not on an instrument but on the exclusion of experienced teachers, the only ones able to choose their posting, with estimation carried out through multilevel models on third-grade pupils. One fewer pupil raises scores by 2.6% of a standard deviation in reading and 2.7% in mathematics, and up to 8% in priority education zones.

Bressoux and Lima (2011) re-examine, using the same method, the 2002–2003 “reduced-size first grade” experiment, which was neither randomised nor located anywhere outside priority education. They show that the official evaluation was biased downwards: the treatment coefficient rises from 0.136 to 0.216 once teacher seniority is controlled for, i.e. 2.2% of a standard deviation per pupil, which the authors regard as a lower bound.

2 Additional studies to the review by Bouguen, Grenet, and Gurgand (2017)

2.1 Instrumental variables studies

2.1.1 Angrist et al. (2019)

Twenty years after their foundational 1999 study, the same authors reproduce exactly the same identification strategy on a far larger sample: 240,310 fifth-grade pupils tested between 2002 and 2011. The estimates are not significantly different from zero, in contradiction with the -0.25 to -0.28 of the original paper. This result is all the more striking in that the authors show it cannot be explained by manipulation of enrolment around Maimonides’ thresholds, which is nonetheless well documented in the recent data. We discuss this result at greater length after the replication conducted in Part III.

2.1.2 Angrist, Battistin & Vuri (2017)

Angrist, Battistin and Vuri (2017) apply Angrist and Lavy’s (1999) identification strategy to Italian primary schools, exploiting the regulatory class-opening thresholds (25 pupils before 2009, 27 afterwards) covering roughly 140,000 second- and fifth-grade classes between 2009 and 2012. On the face of it, their first 2SLS estimates reproduce the expected pattern: reducing class size improves scores, with an effect concentrated in the Mezzogiorno, two to three times larger than that observed in the rest of the country.

The authors nonetheless cast doubt on the validity of this result by exploiting a second quasi-experimental device, independent of the threshold rule: the random assignment of external monitors to about 20% of Italian schools, designed to limit teacher cheating during the manual transcription of pupils’ answers onto marking sheets. They show that this external monitoring sharply reduces the share of classes with manifestly falsified scores. The phenomenon affects up to 14% of classes in the South against 5% nationally.

By estimating a model in which class size and score manipulation are treated simultaneously as two endogenous variables, each separately instrumented (the first by the threshold rule, the second by the random assignment of monitors), the authors show that the effect of class size on scores becomes statistically null in every region, including the South, once manipulation is neutralised, while the effect of manipulation itself remains massive and highly significant. In other words, what the usual 2SLS estimate attributed to a learning gain in the small classes of the Mezzogiorno stems entirely from a greater propensity among teachers to rig results, and not from any causal effect of class size on pupils' actual learning.

2.1.3 Shen & Konstantopoulos (2021)

Shen and Konstantopoulos (2021) study class size effects in secondary education in 2003, 2007 and 2011, focusing on eighth-grade pupils in four European countries: Hungary, Lithuania, Romania and Slovenia. They use the instrumental variables method based on national maximum-enrolment rules as well as regression discontinuity designs, and examine both cognitive outcomes (scores in mathematics, physics, biology, chemistry and earth sciences) and non-cognitive ones (engagement, motivation, self-perception). The estimates reveal positive and significant effects of class-size reduction in Romania in 2003 on scores in several scientific subjects, as well as effects on certain non-cognitive dimensions in Lithuania. These effects, where significant, are of considerable magnitude, larger than those measured in the STAR experiment. The authors stress that class size effects depend heavily on national context and grade level, and that detecting them is more likely in countries with limited school resources.

2.1.4 Urquiola (2006)

Urquiola (2006) studies rural Bolivia, a developing country. The study rests on two distinct identification strategies, only one of which is an instrumental variables approach.

The first is not an IV strategy. It exploits rural schools with 30 or fewer pupils per cohort, which have only one class per grade. Class size there is mechanically equal to cohort enrolment. On this subsample, the correlations between enrolment and socio-economic status become markedly weaker than in the full sample, reducing the scope for bias. On this subsample, removing one pupil raises marks by about 0.22 points in language and 0.19 points in mathematics. Relative to the standard deviation of the rural score (9.3 in language, 9.7 in mathematics), this represents a gain of about 2.4% of a standard deviation per pupil in language and 2.0% in mathematics.

The second strategy uses the IV approach, reproducing the logic of Maimonides' rule. Bolivian regulation allows schools whose enrolment exceeds 30 pupils in a given grade to request an additional teacher, which generates a discontinuity in the relationship between enrolment and class size. Unlike the Israeli case, this rule is not applied mechanically: authorities often wait for enrolment to reach 38 before appointing a new teacher, which produces a fuzzy rather than a sharp discontinuity. Urquiola nonetheless uses the class size predicted by the rule as an instrument in a 2SLS procedure restricted to rural schools. This IV strategy produces somewhat larger per-pupil effects: depending on the specification, the gain ranges from 1.9% to 3.5% of a standard deviation per pupil in language and from 2.2% to 2.6% in mathematics. Thus, expressed per pupil, the two designs remain close.

2.2 Evaluating the French class-size halving policy: Andreu et al. / DEPP (2021) and El Bah / DEPP (2026)

The evaluation of the French class-size halving policy rests on two DEPP studies, which measure its short- and long-run effects, both using difference-in-differences. This method compares the progress of the treated group between before and after with that of an untreated control group, and attributes the gap between the two to the reform: the control group's progress serves as the counterfactual, capturing the effect of time common to both groups. Its validity rests on an assumption, known as parallel trends: absent the reform, both groups would have followed the same trajectory. Levels may differ, but trajectories must coincide. Bearing on an unrealised state of the world, this assumption cannot be verified directly; it can only be made credible by observing the years preceding treatment.

Andreu et al. (2021). The study evaluates the halving of first- and second-grade classes in REP+ schools for the cohort entering first grade in 2017, followed to the end of second grade. REP+ pupils are compared with a control group termed “Near REP+”, made up of 104 REP schools selected by matching on social and educational characteristics close to those of REP+ schools. Over the first-to-second-grade period, REP+ pupils progress more: the modelled effect is 9 % of a standard deviation in French and 14 % in mathematics, which, relative to the seven-pupil gap between the two groups, amounts to 1.3 % and 2 % of a standard deviation per pupil removed. Estimated on the first-grade year alone, the same model gives +14 % in French and +12 % in mathematics. In mathematics the two-year effect is therefore essentially acquired during first grade; in French the first-grade gain exceeds the two-year total, which points to an erosion during second grade rather than to a mere absence of further progress, a reading the authors themselves adopt.

The authors also examine whether the effect differs by pupil attainment level, which bears directly on a policy targeted at disadvantaged backgrounds. At each observation point and in each subject, they determine the score below which the weakest 20 % of control-group pupils fall, then apply this threshold to REP+ schools: a share above 20 % signals more frequent difficulties than in Near REP+ schools, a share below 20 % less frequent ones. In mathematics, the share of REP+ pupils below this threshold falls from 21.4 % at the start of first grade to 15.9 % at the end of second grade, while the share of top-performing pupils remains stable; in French, no comparable movement appears, the share going from 22.4 % to 22.3 %. Relative to the standard deviation of the indicator for belonging to the struggling group, i.e. $\sqrt{0.2 \times 0.8} = 0.4$, the 5.5-point fall corresponds to about 14 % of a standard deviation, which the authors compare with the 8 % descriptive effect on mean scores. The effect thus appears stronger at the bottom of the distribution than at the mean.

Three reservations are in order. First, the parallel trends assumption is not tested: built around the single cohort that inaugurates the reform, the study has no pre-treatment year available, and comparability rests entirely on matching initial levels. Second, the measured effect is composite: in REP+ schools the class-size reduction was accompanied by continuing teacher training and enhanced supervision, which difference-in-differences cannot separate from the reduction in class size alone; the 1.3 % figure is therefore an upper bound. Third, the quintile analysis remains descriptive: no difference-in-differences model is estimated on this indicator, and the conversion into 14 % of a standard deviation is a back-of-the-envelope

calculation, presented in a footnote. The authors themselves write that the improvement “appears” stronger for struggling pupils.

El Bah / DEPP (2026). The study extends this evaluation to entry into lower secondary school, drawing on the national assessments taken at the start of *sixième*, the first year of French lower secondary school (age 11), from 2017 to 2024. It retains the same control group as in 2021, REP schools, treated only from 2018, but this time tests the parallel trends assumption, which the depth of the data makes possible: these assessments have existed since 2017, whereas the first halved REP+ cohort only reaches that point in 2022, leaving five pre-reform years to check the stability of the gap between the two groups. The test takes the form of a regression on interactions between the treated group and each pre-reform year: if the gap was stable before the reform, these coefficients should be null; a significant coefficient would reveal a pre-existing divergence, which would contaminate the estimate.

The test reaches opposite conclusions depending on the comparison. For REP+ versus REP, the pre-reform coefficients are non-significant on scores, with the exception of 2020 in mathematics, attributable to the pandemic. For REP versus non-priority education, most are significant: the trajectories were already diverging, with non-priority schools serving a population too heterogeneous to act as a counterfactual for REP schools.

The result is null: for the first treated cohort, entering lower secondary school in 2022, the difference-in-differences coefficient is -0.007 standard deviations in French and $+0.004$ in mathematics, both non-significant. No effect appears on self-confidence either. Only an analysis restricted to the French overseas departments, for which parallel trends also hold, reveals a weak but significant positive effect ($+0.051$ standard deviations) in French.

The study does not revisit the 2021 quintile split, but examines in an appendix a second bottom-of-distribution indicator: the rate of grade retention on entry to lower secondary school, i.e. the share of pupils at least one year behind. The estimate suggests a decrease of 0.5 percentage points, significant at the 1% level, for cohorts from REP+ schools. This result, unlike those on scores, cannot however be interpreted causally: the parallel trends test reveals several significant pre-reform coefficients, which the authors describe as a clear violation of the assumption.

2.3 Synthesis and comparative table

Studies using credible identification strategies most often find negative and significant effects, generally on the order of 1.7 to 2.5% of a standard deviation per pupil removed in primary school, magnitudes to be considered substantial in light of Kraft’s (2020) benchmarks.

These estimates are not, however, directly comparable. The studies listed in Table A.1, in the appendix, all ask whether smaller classes help pupils learn, but they exploit variations of very different kinds: a continuous shift induced by Maimonides’ rule in Angrist and Lavy (1999), a cut of some eight pupils under randomisation in Krueger (1999), a cut of seven pupils at a stroke under the REP+ reform in Andreu et al. (2021). We therefore bring them onto a common scale, the gain associated with removing a single pupil, measured as a percentage of the standard deviation of class mean scores, by dividing each total effect by the corresponding change in class size. For the DEPP studies, this gives $9\% / 7 = 1.3\%$ of a standard deviation

per pupil in French in first grade.

Two reservations attach to this figure. The first is specific to that study: resting on groups that were not initially equivalent, without a parallel trends test, and measuring an effect that bundles class-size reduction with the training and supervision measures accompanying it, 1.3% is an upper bound on what the reduction alone achieves.

The second bears on the rescaling itself, and therefore on the whole table. Dividing by the number of pupils removed assumes the effect is linear in class size, that is, that going from 40 to 39 pupils yields the same gain as going from 20 to 19. Nothing guarantees this, and the assumption is stretched furthest where it matters most: the variations being compared run from seven pupils in REP+ schools to some twenty at Maimonides' thresholds. The table therefore supports a comparison of orders of magnitude, not a ranking.

Part II. Analysis and replication of Angrist and Lavy (1999): methodological framework

1 The endogeneity problem

The structural equation of interest is the class-level aggregate model proposed by Angrist and Lavy (1999): the class mean score $\bar{y}_{sc}$ depends on observable characteristics X_{sc} , actual class size n_{sc} , a school-specific effect μ_s and a class-specific error term ε_{sc} .

$$\bar{y}_{sc} = X'_{sc}\beta + n_{sc}\alpha + \underbrace{\mu_s}_{\text{common to school } s} + \underbrace{\varepsilon_{sc}}_{\text{specific to class } c}$$

The parameter of interest is α , which measures the expected causal effect of one additional pupil on the class mean score. The school random effect μ_s captures unobserved factors common to all classes in a given school, such as the quality of the leadership, the school culture or the school climate. By contrast, the error term ε_{sc} captures shocks or unobserved factors specific to a particular class, such as an individual teacher, group dynamics or measurement noise. The essential difference is therefore that μ_s is shared by all classes in school s , whereas ε_{sc} varies from one class c to another.

We want to estimate α , the causal effect of one additional pupil on scores. But n_{sc} is endogenous: advantaged families choose small classes, principals place weaker pupils in small classes, and so on. These mechanisms create a correlation between n_{sc} and ε_{sc} . OLS captures this selection effect on top of the causal effect, so its estimator is biased.

2 The instrumental variables estimator: theory and the 2SLS procedure

2.1 Validity conditions

To address the endogeneity problem, Angrist therefore turns to an instrument. A valid instrument Z must satisfy two conditions. First, relevance: Z must be correlated with the endogenous variable n_{sc} , which is verified empirically by the significance of the first stage. Second, the exclusion restriction: Z must affect the score $\bar{y}_{sc}$ only through n_{sc} . This condition cannot be tested formally; it must be argued theoretically.

2.2 The 2SLS procedure

The two-stage least squares (2SLS) procedure proceeds in two steps. In the first stage, actual class size is regressed on the instrument and the controls:

$$n_{sc} = \delta_0 + \delta_1 f_{sc} + \gamma_1 SES_s + \gamma_2 e_s + u_{sc}$$

Estimating this equation yields two things of a different nature, which it is important to distinguish. The *coefficients* $\hat{\delta}_0$, $\hat{\delta}_1$, $\hat{\gamma}_1$ and $\hat{\gamma}_2$ are numbers, unique across the whole sample: $\hat{\delta}_1$ in particular indicates by how much actual size increases when the size predicted by the rule increases by one pupil. The *fitted value* $\hat{n}_{sc}$, by contrast, is a variable: obtained by applying these coefficients to each class's characteristics, it takes a different value for each, since each has its own instrument, its own social composition and its own cohort enrolment. It is a fictitious class size, the one the class would have if its size were entirely determined by the rule and the controls.

The first stage thus decomposes observed actual size into two components, $n_{sc} = \hat{n}_{sc} + \hat{u}_{sc}$. The first retains only what comes from the mechanical discontinuities of the rule and from the controlled variables. The second, the residual, contains the endogenous variation: the principal's decisions, families' choices, selection effects. By keeping only $\hat{n}_{sc}$, we ensure that the variation exploited in the second stage is exogenous by construction.

In the second stage, the mean score is regressed on this fitted variable, in place of actual size:

$$\bar{y}_{sc} = \beta + \alpha \hat{n}_{sc} + \gamma_1 SES_s + \gamma_2 e_s + \varepsilon_{sc}$$

Since $\hat{n}_{sc}$ is no longer correlated with the error term ε_{sc} , the least squares estimator applied to this second equation is no longer biased, and $\hat{\alpha}_{IV}$ measures the causal effect of one additional pupil on the class mean score. In practice, the estimates are carried out via the `IV2SLS` function of the `linearmodels` package, which chains the two stages together and computes correct standard errors.

When there is a single instrument, as here, these two stages admit an equivalent and more telling formulation, the Wald ratio:

$$\hat{\alpha}_{IV} = \frac{\hat{\pi}_1}{\hat{\delta}_1}$$

where $\hat{\pi}_1$ is the coefficient from a direct regression of the score on the instrument, known as the reduced form. This is not an additional step in the procedure: it reaches the same result by another route. The intuition is simple. One additional pupil in the size predicted by the rule raises actual class size by $\hat{\delta}_1$ pupils and changes the score by $\hat{\pi}_1$ points; the effect of one actual pupil is obtained by taking the ratio of the second to the first. It is this formulation that we use in Part III to set out the estimation before carrying it out with the IV2SLS function.

3 Maimonides' rule as an instrument

The class size predicted by Maimonides' rule is:

$$\hat{f}_{sc} = \frac{e_s}{\left\lfloor \frac{e_s - 1}{40} \right\rfloor + 1} \quad (1)$$

where e_s denotes the cohort enrolment of school s , that is, the total number of pupils enrolled in the grade considered.

This formula generates a sawtooth pattern, illustrated in Figure 1. Between thresholds, $\hat{f}_{sc}$ increases linearly with e_s : the larger the cohort, the larger the classes. At the thresholds of 40, 80 and 120 pupils, $\hat{f}_{sc}$ drops abruptly: a cohort of 40 pupils gives $\hat{f}_{sc} = 40$, a single class, whereas a cohort of 41 pupils gives $\hat{f}_{sc} = 20.5$, i.e. two classes. This discontinuity is purely mechanical: it depends only on cohort enrolment, fixed by school registrations, and not on decisions taken by principals, families or education authorities.

The relevance condition requires that $\hat{f}_{sc}$ be correlated with observed class size. This is the only one of the two conditions that is directly verifiable, and it is examined in Table A.6. The exclusion restriction, by contrast, cannot be tested directly. It rests on the following idea. $\hat{f}_{sc}$ is a deterministic function of e_s : it therefore inherits only what is correlated with cohort size. Now, if an omitted determinant of scores varies with e_s , it varies with it *smoothly*, a cohort of 41 pupils not differing fundamentally from a cohort of 39, whereas $\hat{f}_{sc}$ *jumps*. It is this difference in regularity that identifies the effect.

3.1 Why controlling for e_s is necessary

The instrument $\hat{f}_{sc}$ solves the endogeneity problem of n_{sc} by exploiting only the mechanical variation in class size. Since it depends on no human decision, the omitted variables that move class size and mean scores simultaneously, such as families' choices or the principal's decisions, can no longer bias the estimate. That channel is short-circuited by construction.

What remains is the mechanical link between $\hat{f}_{sc}$ and cohort enrolment, which opens a spurious channel. In Israel, large cohorts are found disproportionately in large cities, where pupils are on average more advantaged. An omitted variable Z , neighbourhood affluence for instance, may therefore attract families, increase e_s , and improve scores without passing through class

size: $Z \rightarrow e_s \rightarrow \bar{y}_{sc}$. Since $\hat{f}_{sc}$ is a function of e_s , it inherits this channel.

It is worth stating exactly what this channel can transmit. Decompose $Z = E[Z | e_s] + u$, where u is the part of Z that varies across schools with the same cohort enrolment. Since $\hat{f}_{sc}$ is entirely determined by e_s , u is orthogonal to it by construction and cannot bias anything, whatever its correlation with scores. Only $E[Z | e_s]$ is problematic. It is therefore not necessary to observe Z , nor even to know that it exists: it suffices to model its relationship with e_s well. That is what controlling for e_s does.

This control, however, can only be parametric. Conditioning fully on e_s , through cohort fixed effects, would eliminate all variation in $\hat{f}_{sc}$ and make estimation impossible: the first stage would be null by construction. We therefore remove only the smooth component of e_s , in linear, logarithmic or quadratic form, identifying the effect from what remains, namely the jumps at the thresholds.

The identifying assumption therefore depends on how the enrolment channel is neutralised. With a parametric control on the full sample, identification exploits the whole sawtooth pattern and not only its jumps: the assumption is then that the chosen functional form approximates $E[Z | e_s]$ well, which Table 1 will show to be a demanding requirement. It reduces to the weaker requirement that $E[Z | e_s]$ be continuous at the thresholds, while $\hat{f}_{sc}$ jumps there, only in the discontinuity sample of Section 4.1. This weaker assumption is plausible because the thresholds arise from an administrative counting rule with no social or economic meaning: there is no reason why an omitted variable should shift abruptly at exactly 40, 80 and 120 pupils. Figure 1 illustrates this: the vertical lines mark the thresholds, and while predicted size drops there by twenty pupils, the scatter of the share of disadvantaged pupils crosses them without a break.

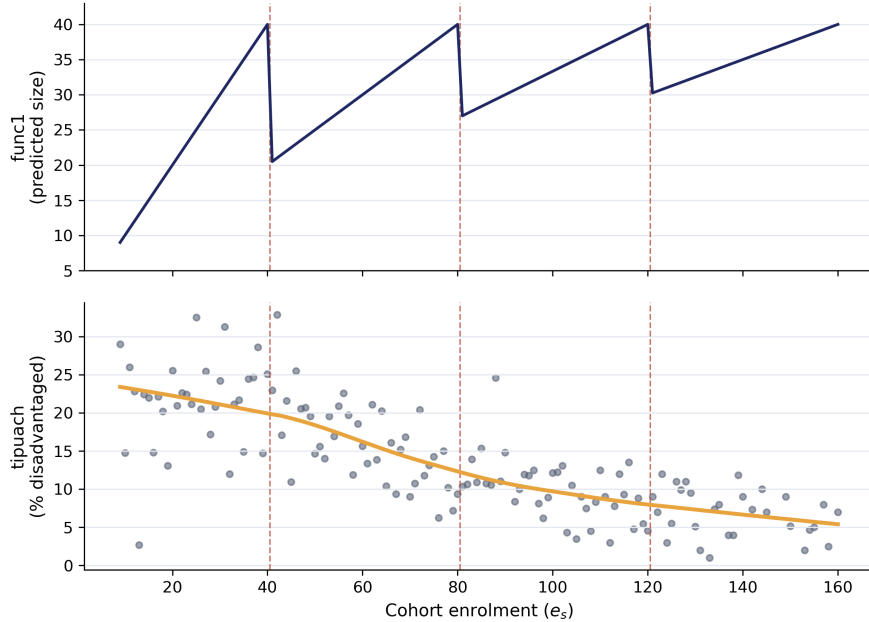


Figure 1: Top: predicted size $\hat{f}_{sc}$ by cohort enrolment value: the instrument drops abruptly at the thresholds, marked by the vertical lines. Bottom: the share of disadvantaged pupils: social composition declines with cohort size, but smoothly, with no break at the thresholds. The decline is concave, which explains why a linear control in e_s does not suffice to absorb it.

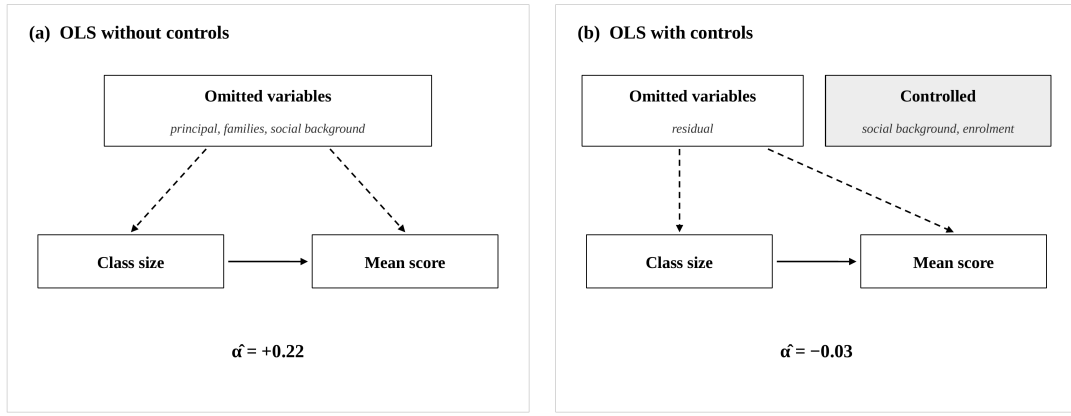


Figure 2: Selection bias in ordinary least squares. Solid lines represent structural relationships, dashed lines the bias channels. Adding observable controls reduces the bias without eliminating it, as other omitted variables remain correlated with classsize.

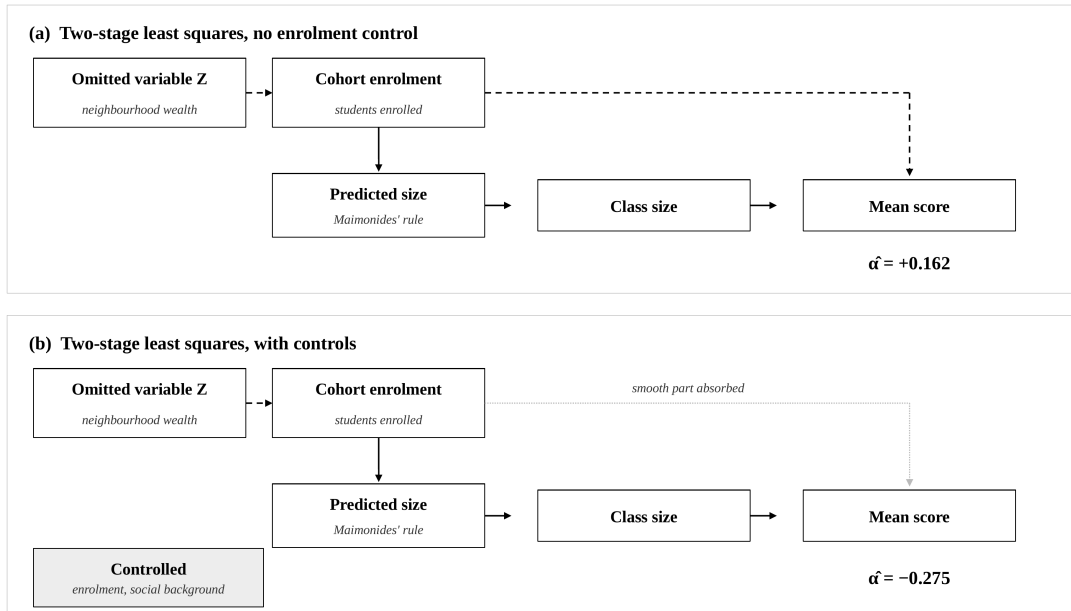


Figure 3: The instrument inherits the endogeneity of e_s because it is a deterministic function of it. Without controls, the channel $e_s \rightarrow avgverb$ violates the exclusion restriction. The parametric control absorbs its smooth component, the dotted line recalling that the absorption is not complete.

Figures 2 and 3 summarise this reasoning in four situations. With neither instrument nor controls, ordinary least squares fully captures the selection bias and produces a spuriously positive coefficient. Adding observable controls reduces this bias without eliminating it, since other omitted variables remain correlated with class size. Moving to 2SLS cuts the channel of classic omitted variables, but leaves the channel $e_s \rightarrow \bar{y}_{sc}$ intact, which must therefore be dealt with separately.

This is where the direct control for the school's social composition comes in. Controlling for e_s should in principle suffice, since this channel can reach scores only by passing through

cohort enrolment. But it neutralises it only to the extent that the chosen functional form approximates it, and that form is unknown: nothing indicates that the relationship between social composition and cohort size is linear, logarithmic or quadratic. Introducing the share of disadvantaged pupils directly makes it possible to address the most important channel without having to make that assumption. Angrist and Lavy do retain this control in all their specifications. The combination of the instrument, the control for e_s and the control for social composition thus constitutes the main specification. Section 4 of Part III tests its sensitivity to the choice of functional form, comparing linear, logarithmic and quadratic forms.

3.2 Standard errors

Standard errors are corrected for within-school correlation. Classes in the same school share common characteristics (the same principal, the same neighbourhood, the same resources) that induce a positive correlation between their residuals. If this correlation is ignored and all classes are treated as independent, standard errors are underestimated and inference becomes overly optimistic.

Angrist and Lavy apply the analytical correction of Moulton (1986), implemented in their Stata program via the `mmoulton` command. The idea is to compute explicitly ρ_e , the intraclass correlation of the residuals, that is, how much the residuals of two classes in the same school resemble one another, and then to inflate the OLS standard errors by a factor that depends on this correlation and on the average group size:

$$SE_{\text{Moulton}} = SE_{\text{OLS}} \times \sqrt{1 + \rho_e \rho_x \left(\frac{\sigma_m^2}{\bar{m}} + \bar{m} - 1 \right)}.$$

This formula is analytical and closed-form. It assumes that the correlation structure is homogeneous across schools, that is, that all schools resemble one another in the same way. It reduces t statistics and makes inference more conservative.

In our Python replication, we adopt a different but asymptotically equivalent approach: the *cluster-robust* estimator. Rather than computing ρ_e analytically, we aggregate residuals directly school by school and allow their correlation structure to be arbitrary.

3.3 LATE

The IV estimator identifies a local effect, the LATE (*Local Average Treatment Effect*), that is, the causal effect for schools whose class size actually varies in response to Maimonides' thresholds. These schools, called *compliers*, are those whose cohort enrolment lies close to the thresholds of 40, 80 and 120 pupils. Schools whose enrolment is far from the thresholds do contribute, since $\hat{f}_{sc}$ also varies with e_s within each segment, but they weigh far less in the estimate: the variation in predicted size they generate is gradual, whereas it reaches some twenty pupils at the thresholds. The 2SLS estimator weights each school by the strength of its response to the instrument, so schools near the thresholds dominate identification. The LATE may therefore differ from the average treatment effect (ATE) that would be obtained in a randomised experiment such as STAR, if the effect of class size is heterogeneous across school characteristics. This is a limitation inherent to any instrumental variables identification

strategy.

Part III. Replication of Angrist & Lavy (1999) in Python

Preliminary note. Tables reproducing Angrist and Lavy’s results are reported in the appendix; those belonging to our extension appear in the main text.

1 Data and descriptive statistics

The data come from the national testing programme in Israeli primary schools, June 1991. The unit of observation is the class. The files `final5.dta`, for fifth-grade classes, and `final4.dta`, for fourth-grade classes, made available by Angrist on his institutional website, contain respectively 2,029 and 2,059 classes before any processing.

The main variables are actual class size, `classsize`, cohort enrolment, `c_size`, mean reading scores, `avgverb`, and mathematics scores, `avgmath`, and the share of disadvantaged pupils, `tipuach`, which serves as the socio-economic control (denoted SES in Angrist and Lavy’s paper). To these is added `func1`, the class size predicted by Maimonides’ rule, which we construct according to equation (1) and which bears this name in the authors’ original program. Table A.2, in the appendix, presents descriptive statistics for the raw fifth-grade data, before any filters are applied.

Before estimating the regressions, the raw data are subjected to several corrections and filters, identical to those of Angrist and Lavy’s original Stata program.

One class has an `avgverb` score of 187.6, an impossible value for a score meant to lie between 0 and 100. The original Stata program corrects this score by subtracting 100, with no justification documented in the paper or in the data appendix. We adopt a more cautious approach, excluding this observation, though the choice is inconsequential since it concerns a single class.

Five further classes are excluded because no pupil sat the reading test, `verbsize` = 0, making the `avgverb` mean incomputable.

Three filters are then applied. Classes with fewer than or equal to 1 pupil or greater than or equal to 45 pupils are excluded as atypical classes or data-entry errors. Cohorts of 5 pupils or fewer are also discarded, since Maimonides’ rule has no empirical meaning for such small enrolments. Finally, the sample is restricted to public schools in the non-religious Jewish sector (`c_leom` = 1 and `c_pik` < 3), in line with the scope of the original study. The Israeli school system distinguishes the Jewish sector from the Arab sector and, within the former, general public education, religious public education and independent ultra-Orthodox education; only the first is retained.

After applying these filters, the final sample contains 2,018 fifth-grade classes and 2,049 fourth-grade classes, a sample very close to the original one. These filters therefore do not bias the data. Tables A.3 and A.4, in the appendix, present descriptive statistics for this final

sample, corresponding to Table I of the original paper.

All the estimates that follow rest on two utility functions, `ols_cluster` and `iv_cluster`, which apply the school-level standard error correction discussed in Section 3.2; their implementation is detailed in the appendix.

2 OLS results

Table A.5, in the appendix, presents the OLS results of the first naive regressions. We recover the same results as Table II of Angrist and Lavy. It confirms the OLS bias and illustrates the endogeneity problem described in Part II.

Without controlling for `tipuach`, columns 1 and 4, the coefficient on `classsize` is positive and highly significant: +0.221 for reading and +0.322 for mathematics. This counter-intuitive result means that, on the face of it, larger classes are associated with better scores. This is selection bias at work: large classes are found disproportionately in large Israeli cities, where pupils are on average more advantaged.

As soon as `tipuach` is controlled for, columns 2 and 5, the sign flips: the coefficient becomes negative for reading, -0.031 , and falls sharply for mathematics, $+0.076$ against $+0.322$ without the control. Adding cohort enrolment e_s , columns 3 and 6, renders the coefficients non-significant in both subjects.

But columns 3 and 6 do not solve the endogeneity problem: they absorb part of the selection bias through observable controls, while `classsize` remains correlated with the error term through unobservable mechanisms, such as the principal's decisions, non-random grouping of pupils within a school, or families' requests. It is for this reason that Angrist and Lavy turn to Maimonides' rule as an exogenous instrument.

3 First stage and reduced form

Before estimating 2SLS, Table A.6, in the appendix, documents two intermediate steps that underpin its validity and allow its intuition to be set out.

The first panel presents the first stage, i.e. the regression of actual class size (`classsize`) on the size predicted by Maimonides' rule (`func1`, i.e. the instrument). Without controlling for cohort enrolment, column (1), the coefficient is 0.70: too high, because the instrument still picks up the direct effect of cohort size on class size. As soon as cohort enrolment is controlled for, column (2), it falls to 0.54, which matches the paper's value exactly. This coefficient means that one additional pupil predicted by Maimonides' rule translates into an increase of 0.54 pupils in actual class size.

The second panel presents the reduced form for reading, i.e. the direct regression of the mean reading score (`avgverb`) on predicted size (`func1`). Without controlling for cohort enrolment, column (3), the coefficient is -0.111 . With the control, column (4), it is -0.149 , exactly the paper's value. This coefficient measures the effect of the instrument on scores, without going through 2SLS.

The third panel presents the reduced form for mathematics. Without controlling for cohort enrolment, column (5), the effect is near zero and non-significant. With the control, column (6), it becomes -0.124 and significant at the 5% level. This more fragile result foreshadows that the IV effects on mathematics will be less robust than for reading.

The relevance condition stated in Section 3 is therefore amply satisfied: the associated t statistic exceeds 14, well beyond conventional weak-instrument thresholds, and the first stage yields an R^2 of 0.51. The instrument thus explains, together with the controls, about half the variation in actual class size.

The Wald ratios follow directly from these two panels and constitute the simplest IV estimator with a single instrument, as set out in Part II. For reading:

$$\frac{-0.149}{0.542} = -0.275.$$

For mathematics:

$$\frac{-0.124}{0.542} = -0.229.$$

These ratios exactly foreshadow the results in Table A.7, discussed in the next section.

4 2SLS results

The Wald ratios in the previous section served to set out the intuition of the IV estimator: two separate OLS regressions, a first stage and a reduced form, whose coefficients are divided. Table A.7, in the appendix, presents the same results via the `IV2SLS` function of the `linearmodels` package, which encapsulates the two stages in a single command and makes it easy to test several specifications of the control for e_s , with the associated standard errors, t statistics and p -values. All columns include the socio-economic control `tipuach`; they differ only in their treatment of cohort enrolment.

Column (2) is estimated without a control for e_s . The IV coefficient on reading is -0.158 , significant at the 1% level. It is however biased towards zero because the instrument still picks up part of the direct effect of cohort size on scores, as seen in Table A.6.

Column (3), with a linear control for e_s , is the main specification. The IV coefficient is -0.275 for reading and -0.229 for mathematics. Both coefficients are statistically significant and correspond to the Wald ratios computed from Table A.6. One additional pupil in the class therefore reduces the mean reading score by 0.275 points and the mathematics score by 0.229 points.

Columns (4) and (5) test the robustness of this result by replacing the linear control with a logarithmic and then a quadratic form of e_s . This test concerns the direct effect of cohort size on scores, whose form is not known a priori: the channel running through social composition is already handled by `tipuach`, but other channels may link cohort enrolment and outcomes, school resources or peer effects, with no particular reason to be linear. The choice of specification does not, however, drive the estimate: the coefficients remain negative and significant in both subjects, and the values obtained for reading, -0.304 and -0.259 , bracket the main estimate of -0.275 . The specification retained is the linear form, for comparability with column (2) of Angrist and Lavy's (1999) Table IV, but the conclusion does not depend on this choice.

One question remains that reading Table A.7 cannot settle, since `tipuach` appears everywhere in it: would a sufficiently flexible control for e_s not render that control superfluous? Social composition indeed reaches scores only by passing through cohort enrolment, so that in principle, modelling that relationship well should suffice. Table 1 shows that this is not the case.

Table 1: Is the socio-economic control redundant with the control for e_s ? (reading, fifth grade)

Form of e_s	Without <code>tipuach</code>	With <code>tipuach</code>
Linear	−0.115 (0.093)	−0.275*** (0.076)
Quadratic	−0.176 (0.118)	−0.259*** (0.094)
Logarithmic	−0.185 (0.126)	−0.304*** (0.101)

Notes: 2SLS coefficient on `classsize`, dependent variable `avgverb`, instrument `func1`. Cluster-robust standard errors at the school level in parentheses. The right-hand column reproduces columns (3), (5) and (4) of Table A.7. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The socio-economic control therefore cannot be replaced by a more flexible treatment of enrolment. Without it, the estimate falls to −0.115, and the quadratic and logarithmic forms, which better approximate the concave relationship between social composition and cohort enrolment, bring it back only to −0.176 and −0.185: none attains significance. The parametric control in e_s therefore lets a substantial part of this channel through, which `tipuach` addresses directly, with no assumption about the form of the relationship. It also improves precision, with standard errors falling by about 20%, social composition explaining an important share of score variance even across schools with the same cohort enrolment.

This finding extends beyond social composition, and constitutes the main limitation of this strategy on the full sample. Nothing guarantees that some unobserved variable does not likewise vary with cohort enrolment in a way that the parametric control approximates poorly, and Table 1 shows precisely that such a case can shift the estimate by half. This is what motivates the discontinuity sample presented in Section 4.1: by restricting the analysis to a narrow window around the thresholds, it renders any smooth function of e_s quasi-constant and thereby dispenses with any functional-form assumption.

In terms of *effect size*, the main result for reading corresponds to a gain of about 0.36 standard deviations for a 35% reduction in class size, i.e. moving from a class of 30 pupils to one of 20. This figure is comparable to the effects measured by the STAR experiment in Tennessee, which strengthens the external credibility of Angrist and Lavy’s results.

Panel B of Table A.7 presents the same specifications for fourth-grade pupils. The results are noticeably weaker and less robust. For reading, the IV coefficient is −0.133 in the main specification, significant but roughly half as large as in fifth grade. For mathematics, no specification produces a significant coefficient. These results suggest that the effect of class size is more pronounced for younger pupils, or that the difference between fourth and fifth grade reflects a cumulative effect: fifth-grade pupils have been exposed longer to large classes, which makes variation in size more decisive for their learning.

4.1 Robustness: discontinuity sample and binary instruments

The preceding estimates exploit the continuous variation of $\hat{f}_{sc}$ across the whole sample. For this strategy to be valid, one must control for e_s : since $\hat{f}_{sc} = f(e_s)$, the instrument mechanically inherits everything that influences cohort enrolment. The linear, logarithmic and quadratic specifications of Table A.7 all seek to absorb this direct effect, but each rests on an arbitrary functional assumption: if the true effect of e_s on scores is neither linear, nor logarithmic, nor quadratic, a residual bias persists.

Rather than modelling this relationship, we therefore restrict the sample to cohorts falling within a narrow window of ± 5 pupils around each Maimonides threshold, i.e. the intervals [36–45], [76–85] and [116–125]. Within a window of ten values, e_s varies so little that its effect on scores is necessarily negligible, whatever its functional form. The discontinuity sample comprises 471 fifth-grade classes and 415 fourth-grade classes, i.e. about 23% and 20% of the full samples.

Identification then rests on two sets of indicator variables that must be distinguished. The controls are two window indicators, seg_1 for the interval [36–45] and seg_2 for [76–85], the third window serving as the reference category; they absorb differences in mean level across intervals, since a school near 40 pupils is not comparable to a school near 120. The instruments are three upper-half indicators, $z_1 = \mathbf{1}[41 \leq e_s \leq 45]$, $z_2 = \mathbf{1}[81 \leq e_s \leq 85]$ and $z_3 = \mathbf{1}[121 \leq e_s \leq 125]$, which equal 1 where the rule predicts the opening of an additional class. The identifying contrast therefore opposes, within each window, schools just above the threshold to those just below. No parametric control for e_s is included.

This strategy is that of columns (5) and (6) of Angrist and Lavy’s (1999) Table IV, which likewise restrict the sample to the neighbourhood of the thresholds. It differs from the so-called *donut* strategy used by Angrist et al. (2019), which by contrast excludes schools immediately around the first threshold: the latter aims to neutralise enrolment manipulation, not functional form, and yields -0.234 against -0.277 on the full sample.

The results are presented in Table A.8, in the appendix. The coefficients are negative and significant for fifth grade in both subjects, both without the socio-economic control (panel A) and with it (panel B). For fourth grade, the reading coefficient is significant in panel A (-0.358^{**}) but loses significance once **tipuach** is controlled for, an instability consistent with the fragility already observed at that grade level in panel B of Table A.7.

The contrast between the two panels is instructive. On the full sample, removing **tipuach** shifted the estimate by half; here, adding it changes the fifth-grade coefficients by only three hundredths, from -0.654 to -0.687 in reading and from -0.568 to -0.596 in mathematics. Once enrolment is neutralised by restriction rather than by modelling, the social control has no bias left to correct, which supports the view that the difficulty encountered on the full sample stemmed from the parametric approximation and not from the instrument itself.

The coefficients are, on the other hand, two to three times larger in absolute value than in Table A.7, reaching -0.687 for fifth-grade reading against -0.275 in the main specification. Two factors contribute to this. The first concerns sample composition: schools near the thresholds are those for which the class-size variation induced by the rule is strongest, i.e. the *compliers* most responsive to the instrument, whose LATE is naturally higher. The second

concerns precision: standard errors are three to four times larger, 0.264 against 0.076, and the 95% confidence interval around -0.687 runs from about -1.20 to -0.17 . It therefore comfortably contains the main estimate. This specification confirms the sign and significance of the effect without relying on any functional assumption, but it is too imprecise to establish its order of magnitude, which is why it constitutes a check rather than the retained estimate.

4.2 Heterogeneity by socio-economic status: the linear interaction

The preceding results concern the average effect of class size across the whole sample. A natural question is whether this effect is homogeneous or varies with pupil profile. Angrist and Lavy (1999) test this hypothesis formally by introducing an interaction term between class size and the share of disadvantaged pupils (**tipuach**).

The estimated model is:

$$\bar{y}_{sc} = \alpha \cdot \text{classsize}_{sc} + \delta \cdot (\text{classsize}_{sc} \times \text{tipuach}_s) + \gamma \cdot \text{tipuach}_s + \beta \cdot e_s + \varepsilon_{sc} \quad (2)$$

The term δ measures whether the marginal effect of one additional pupil is stronger (more negative) in schools with a high share of disadvantaged pupils. The product $\text{classsize} \times \text{tipuach}$ is endogenous, being the product of an endogenous variable and an exogenous one. It is instrumented by $\hat{f}_{sc} \times \text{tipuach}_s$, following Angrist and Lavy's (1999) strategy. The two endogenous variables are instrumented simultaneously by the two instruments via the 2SLS procedure.

The results are presented in Table A.9, in the appendix. For fifth grade, the interaction term is negative and statistically significant in reading ($\hat{\delta} = -0.008^{**}$, $p = 0.037$) and in mathematics ($\hat{\delta} = -0.010^{**}$, $p = 0.026$). This means that one additional pupil reduces scores more in schools with a high share of disadvantaged pupils. By way of illustration, the total effect of one additional pupil in reading amounts to $-0.155 + (-0.008) \times 6 = -0.203$ points for a school in the first tercile of **tipuach** ($\approx 6\%$ disadvantaged pupils, close to P_{33}), against $-0.155 + (-0.008) \times 15 = -0.275$ points for a school at the sixty-seventh percentile ($\approx 15\%$).

For fourth grade, by contrast, no term is significant, which is consistent with panel B of Table A.7. On the pooled fourth- and fifth-grade sample, the interaction term is significant in mathematics ($\hat{\delta} = -0.007^{**}$, $p = 0.038$) and marginally significant in reading ($\hat{\delta} = -0.005^*$, $p = 0.073$).

This reading nonetheless rests on a linear extrapolation that Part IV puts to the test.

Part IV. Extension: the shape of social heterogeneity

The interaction model of Table A.9, in the appendix, concludes that the effect is more pronounced in schools with a high share of disadvantaged pupils. This conclusion nonetheless rests on an assumption the model does not test: that of a *linear* variation of the effect with **tipuach**. A single interaction term constrains the marginal effect to increase, or decrease, at a constant rate over the whole range of the distribution. Nothing guarantees that this is so, and the stakes are not minor: it is precisely by extrapolating this slope that one would be tempted

to predict the gain expected in the most disadvantaged schools, those targeted by the French class-size halving policy.

This extension relaxes the linearity assumption along two independent lines. The first adds a quadratic term to the interaction model and tests curvature directly. The second, non-parametric, splits the sample into terciles of `tipuach` and subjects the equality of the three effects to a formal test, an approach that follows the one by which Piketty and Valdenaire [19] established their own result on social heterogeneity, while supplying the test they lacked.

Both approaches are then applied to a second outcome, absent from the original paper: the failure rate. Angrist and Lavy’s main estimation indeed bears on the class mean score, yet a mean can rise without struggling pupils making progress. The objective of a class-size reduction policy targeted at disadvantaged backgrounds may not be to raise the mean but to reduce the share of struggling pupils; this second measure is more directly relevant to the question at hand, and it will prove, in the most disadvantaged schools, more informative than the mean.

1 Quadratic interaction: testing curvature directly

We add a second interaction term to the model of Table A.9:

$$\begin{aligned} \bar{y}_{sc} = & \alpha \cdot \text{classsize}_{sc} + \delta_1 \cdot (\text{classsize}_{sc} \times \text{tipuach}_s) + \delta_2 \cdot (\text{classsize}_{sc} \times \text{tipuach}_s^2) \\ & + \gamma_1 \cdot \text{tipuach}_s + \gamma_2 \cdot \text{tipuach}_s^2 + \beta \cdot e_s + \varepsilon_{sc} \end{aligned} \quad (3)$$

The three endogenous variables are instrumented simultaneously by $\hat{f}_{sc}$, $\hat{f}_{sc} \times \text{tipuach}$ and $\hat{f}_{sc} \times \text{tipuach}^2$. This extension requires no arbitrary sample split, and tests directly whether the linearity assumption adopted by Angrist and Lavy [2] constitutes a sufficient approximation.

Table 2: Quadratic interaction `classsize` $\times$ `tipuach`² on scores (fifth grade)

	Reading	Mathematics
<code>classsize</code>	−0.129* (0.070)	−0.027 (0.107)
<code>classsize</code> $\times$ <code>tipuach</code>	−0.0235*** (0.0082)	−0.0270** (0.0112)
<code>classsize</code> $\times$ <code>tipuach</code> ²	+0.00051*** (0.00018)	+0.00047** (0.00023)
First-stage F	255/120/142	254/119/142

Notes: 2SLS, fifth grade. Dependent variable: mean score. Endogenous: `classsize`, `classsize` $\times$ `tipuach`, `classsize` $\times$ `tipuach`²; instruments: $\hat{f}_{sc}$, $\hat{f}_{sc} \times \text{tipuach}$, $\hat{f}_{sc} \times \text{tipuach}^2$; controls: `tipuach`, `tipuach`², e_s . A positive and significant δ_2 indicates a bell-shaped relationship between the effect and disadvantage. The three F values correspond to `classsize`, `classsize` $\times$ `tipuach` and `classsize` $\times$ `tipuach`². Cluster-robust standard errors (school). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The results, presented in Table 2, provide evidence in favour of a non-linear relationship: the coefficient on the quadratic interaction is positive and statistically significant in reading ($\hat{\delta}_2 = +0.00051^{***}$, $p < 0.01$) as well as in mathematics ($\hat{\delta}_2 = +0.00047^{**}$, $p < 0.05$).

A positive $\hat{\delta}_2$ means that the (negative) effect of class size first deepens as **tipuach** increases, reaches its maximum intensity, and then attenuates. This turning point can be read directly off the estimated coefficients. The marginal effect of one additional pupil is indeed no longer a single number in this model: it depends on the level of disadvantage,

$$\frac{\partial \bar{y}_{sc}}{\partial \text{classsize}} = \hat{\alpha} + \hat{\delta}_1 t + \hat{\delta}_2 t^2,$$

and this parabola reaches its vertex where its derivative in t vanishes, i.e. $\hat{\delta}_1 + 2\hat{\delta}_2 t = 0$, whence

$$t^* = -\frac{\hat{\delta}_1}{2\hat{\delta}_2} = \frac{0.0235}{2 \times 0.00051} \approx 23 \quad \text{in reading,}$$

and $t^* \approx 29$ in mathematics. The vertex therefore lies, in both subjects, beyond the sample's third quartile: 75% of classes have a **tipuach** less than or equal to 19 (cf. Table A.3).

This turning point must be read with three caveats.

The first is arithmetical. The vertex is the ratio of two estimated coefficients, and its denominator, $2\hat{\delta}_2$, is a very small number carrying a relative uncertainty of about 35% ($\hat{\delta}_2 = 0.00051$ for a standard error of 0.00018). This uncertainty carries over entirely to the ratio: at the bounds of the confidence interval for $\hat{\delta}_2$, the vertex moves from $t^* \approx 14$ to $t^* \approx 73$. The interval therefore spans much of the range of **tipuach** in the sample.

The second concerns specification. A quadratic model does not establish that the relationship is parabolic: it assumes so, and estimates the parabola that best fits the data. Yet every parabola has a vertex, including when the underlying relationship has none. The t^* obtained here may therefore be a by-product of the functional form adopted rather than a threshold read off the data.

A third caveat concerns identification. The model contains three endogenous variables instrumented by three instruments that are strongly correlated with one another, since **tipuach**² is merely a transformation of **tipuach**. Such collinearity can weaken the joint identification of the system in a way that a univariate F statistic does not necessarily detect.

What the model establishes nonetheless remains solid, and suffices for our purposes: with $\hat{\delta}_2$ significant at 1%, the linearity assumption is rejected. A quadratic term significantly improves fit relative to a straight line, which forbids extrapolating the slope estimated in intermediate schools to the highest levels of disadvantage.

Rejecting the straight line does not, however, tell us the true shape of the relationship, and the three reservations above invite us to confront this result with a different identification method. The decomposition of the sample by tercile of **tipuach**, presented below, imposes no functional form and constructs no correlated instruments: it thus addresses all three objections simultaneously.

2 Tercile decomposition: illustration and counter-test

This subsample decomposition is not merely a counter-test of the quadratic model: it also takes up the approach by which Piketty and Valdenaire [19] established their result on social heterogeneity. Rather than testing heterogeneity through an interaction term, Piketty instead re-estimated his instrumental model separately on each pupil subgroup, then compared the coefficients obtained (cf. § 1.2.2). By splitting the sample here according to terciles of `tipuach`, we adopt this splitting strategy. We thus estimate the main 2SLS model separately on three subsamples defined by terciles of `tipuach`: the low group (`tipuach` ≤ 6 , $N = 750$), the middle group ($6 < \text{tipuach} \leq 15$, $N = 622$) and the high group (`tipuach` > 15 , $N = 646$). Unlike the previous model, this approach assumes no functional form: each group is estimated independently, with its own instrument, without constructing interactions correlated with one another.

Table 3: Original extension: 2SLS results by tercile of *tipuach* and instrument strength (fifth grade)

Group	N	Reading	Maths	F (1st stage)	R^2 (1st stage)
Low (<i>tipuach</i> ≤ 6)	750	−0.045 (0.069)	−0.028 (0.127)	135.1	0.627
Middle ($6 < \text{tipuach} \leq 15$)	622	−0.512*** (0.144)	−0.463** (0.198)	55.6	0.463
High (<i>tipuach</i> > 15)	646	−0.244 (0.197)	−0.235 (0.219)	41.9	0.411
Full sample	2018	−0.275*** (0.076)	−0.229** (0.098)	217.8	0.553

Notes: 2SLS estimation, instrument $\hat{f}_{sc}$ (Maimonides’ rule). Controls: *tipuach*, e_s (linear). Cluster-robust standard errors at the school level in parentheses. The tercile cut-offs are $P_{33} = 6$ and $P_{67} = 15$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The F and R^2 columns refer to the first stage (*classsize* $\sim$ *func1*) estimated separately within each subgroup.

The results are presented in Table 3. In the intermediate-disadvantage group, the effect is strong and highly significant: −0.512*** in reading and −0.463** in mathematics, coefficients markedly larger than those for the full sample (−0.275 and −0.229). In the extreme groups, by contrast, the coefficients are negative but non-significant.

One might object that dividing the sample into three weakens the instrument and biases the estimates. This is not the case: the first-stage F remains well above the threshold of 10 in all three groups (135.1, 55.6 and 41.9), which rules out weak-instrument bias.

It does not follow that the three coefficients are estimated with the same precision. Their standard errors vary by a factor of three (0.069, 0.144 and 0.197), even though the terciles are of similar size (750, 622 and 646 classes): sample size is not the only determinant of precision. Instrument strength indeed declines markedly from one group to the next. The first-stage R^2 falls from 0.63 to 0.46 and then 0.41, and the less the instrument explains the variation in `classsize`, the less precise the 2SLS estimator. Clearing the threshold of 10 rules out weak-instrument bias, not imprecision.

The middle tercile is therefore, in reality, less precisely estimated than the low tercile. If it is

the only one to come out significant, this is not because it is better measured, but because its coefficient is markedly larger (0.512 against 0.045 and 0.244), large enough relative to its standard error to stand out from the noise. At the other end, the high tercile combines the disadvantage of a more modest coefficient with the least precise estimation of the three.

But are the three coefficients, which appear different, really so?

The regressions in Table 3 yield three coefficients: -0.045 for the low tercile, -0.512 for the middle tercile, -0.244 for the high tercile. Read at face value, these figures suggest that the effect of class size peaks in the intermediate group.

But an estimated coefficient is never an exact number: it is a value surrounded by a margin of uncertainty, summarised by its standard error, here 0.069, 0.144 and 0.197. If one could draw three new samples of schools and redo the same calculations, one would not obtain exactly the same figures. Translated into 95% confidence intervals, these margins give $[-0.18; +0.09]$, $[-0.79; -0.23]$ and $[-0.63; +0.14]$. The last almost entirely overlaps the other two: on its own, it is compatible with a null effect as well as with an effect larger than that of the middle tercile. It is this overlap that gives rise to doubt; if the three intervals were narrow and clearly separated, reading the coefficients directly would suffice to conclude.

As we have stressed, only one of the three coefficients is individually significant, that of the middle tercile. It would be tempting to conclude directly that the effect differs by social background: it would exist in the intermediate group and nowhere else. This reasoning is nonetheless faulty, because it conflates two distinct questions.

The significance of a coefficient tests $H_0 : \alpha_g = 0$: is the effect null in this group? The test we conduct tests $H_0 : \alpha_l = \alpha_m = \alpha_h$: are the effects equal to one another? Nothing mechanically links the two: one may observe three coefficients all significant yet statistically indistinguishable from one another, or conversely a single significant coefficient without it being possible to assert that it differs from the other two. This is the point stressed by Gelman and Stern [12]: the difference between significant and not significant is not itself statistically significant.

This is precisely where Piketty and Valdenaire [19] stop (cf. § 1.2.2). They compare 0.715 points for pupils of disadvantaged occupational background with 0.274 points for pupils of advantaged background, and conclude that the effect is markedly stronger among the former, without testing whether this gap survives estimation uncertainty, even though the confidence intervals overlap in their paper too. We are in the same situation, and we propose to settle it with a statistical test.

What the Wald test does.

The Wald test answers a single question: is the gap between the coefficients too large to be merely the product of sampling variation? The null hypothesis is that of homogeneity:

$$H_0 : \alpha_{\text{low}} = \alpha_{\text{mid}} = \alpha_{\text{high}}.$$

If the test rejects (low p), the effects genuinely differ by social background. If it does not reject (high p), one cannot rule out that they are in fact identical.

The principle is as follows. The three coefficients are collected into a vector $\hat{\theta} = (\hat{\alpha}_l, \hat{\alpha}_m, \hat{\alpha}_h)'$

and the hypothesis is written as two differences that H_0 assumes to be null:

$$R\theta = 0, \quad R = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}.$$

The statistic compares these estimated differences with their own uncertainty:

$$W = (R\hat{\theta})'[R\hat{V}R']^{-1}(R\hat{\theta}) \stackrel{H_0}{\sim} \chi^2_2,$$

where $\hat{V}$ is the variance-covariance matrix of the three coefficients. Intuitively, W is large when the gaps between coefficients are large *relative* to the precision with which they are estimated.

The term $\hat{V}$ contains, a priori, covariances between the three coefficients. Here, however, these are null, for a simple reason: **tipuach** is a school characteristic and not a class characteristic. The tercile split therefore partitions whole schools. Since our standard errors are clustered at the school level, no cluster straddles two terciles, and the three estimates are statistically independent:

$$\hat{V} = \text{diag}(\widehat{se}_l^2, \widehat{se}_m^2, \widehat{se}_h^2).$$

The test can therefore be computed directly from the coefficients and standard errors of Table 3, without needing to re-estimate anything.

For this reason we retain the three separate regressions as the sole specification. An alternative would have been to re-estimate the three slopes within a single regression in which the controls (**tipuach**, e_s) are constrained to share a common coefficient across the three groups. We do not adopt it: nothing justifies a priori assuming that disadvantage acts identically across three groups defined precisely by disjoint ranges of that very variable, and this constraint would deliver only a gain in precision at the price of an unverifiable assumption.

In reading, homogeneity is rejected: $W = 8.85$, 2 df, $p = 0.012$. The three effects cannot be regarded as identical. In mathematics, homogeneity is not rejected: $W = 3.52$, 2 df, $p = 0.172$; the observed gaps there are compatible with an effect common to all three groups.

Table 4: Test of equality of effects across terciles of **tipuach (fifth grade)**

	Reading			Mathematics		
	Coef.	(SE)	95% CI	Coef.	(SE)	95% CI
Low ($t \leq 6$)	-0.045	(0.069)	[-0.18; +0.09]	-0.028	(0.127)	[-0.28; +0.22]
Middle ($6 < t \leq 15$)	-0.512***	(0.144)	[-0.79; -0.23]	-0.463**	(0.198)	[-0.85; -0.08]
High ($t > 15$)	-0.244	(0.197)	[-0.63; +0.14]	-0.235	(0.219)	[-0.66; +0.19]
<i>Wald test, $H_0 : \alpha_{\text{low}} = \alpha_{\text{mid}} = \alpha_{\text{high}}$</i>						
W statistic	8.85			3.52		
p -value	0.012			0.172		
Conclusion	H_0 rejected			H_0 not rejected		

Notes: coefficients from the three separate 2SLS estimations, reproduced here for readability of the test; t denotes **tipuach**. Wald test with 2 degrees of freedom, computed from the three separate estimations, whose covariances are null (terciles built on a school-level variable, disjoint clusters). Cluster-robust standard errors at the school level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The joint test therefore establishes that at least two of the three effects differ, but it does not say which. To locate this difference, we examine the three pairwise comparisons. The gap between two terciles can be tested directly, since their covariances are null:

$$\text{Var}(\hat{\alpha}_i - \hat{\alpha}_j) = \widehat{se}_i^2 + \widehat{se}_j^2.$$

Conducting three tests instead of one nonetheless requires raising the bar. If the three comparisons were independent, the probability of obtaining no false positive at the 5% level would be 0.95^3 , and that of obtaining at least one $1 - 0.95^3 \approx 14\%$. We therefore apply the Bonferroni correction, which consists in multiplying each p -value by the number of tests conducted, here three, with the result capped at 1. A gap is thus retained as significant only if it would have been so at a threshold three times as demanding.

Table 5: Pairwise comparisons between terciles (fifth grade)

Comparison	Gap	(SE)	z	Raw p	Bonf. p
<i>Reading</i>					
Low vs middle	+0.467	(0.160)	+2.93	0.003	0.010
Middle vs high	-0.268	(0.244)	-1.10	0.271	0.813
Low vs high	+0.199	(0.209)	+0.95	0.342	1.000
<i>Mathematics</i>					
Low vs middle	+0.435	(0.235)	+1.85	0.065	0.194
Middle vs high	-0.228	(0.295)	-0.77	0.440	1.000
Low vs high	+0.207	(0.253)	+0.82	0.414	1.000

Notes: the gap is computed as the coefficient of the first group named minus that of the second; a positive gap therefore means the effect is less pronounced in the first group. The standard error of the gap is obtained by summing variances, the three estimations being independent. The “Bonf. p ” column multiplies the raw p -values by the number of comparisons conducted; values of 1.000 result from the capping. Coefficients and standard errors are those of Table 4.

Only one comparison stands out in reading: that opposing the low tercile to the middle tercile ($p = 0.010$). The high tercile, for its part, does not differ significantly from either of the other two. The rejection of homogeneity therefore rests entirely on the contrast between the least disadvantaged schools, where the effect is near zero, and schools of intermediate disadvantage, where it is strong. In mathematics, no comparison is significant, in line with the non-rejection of the joint test.

Scope and limitations.

The rejection of homogeneity in reading converges with the quadratic specification of the previous section, where the term `classsize` $\times$ `tipuach`² came out positive and significant at 1%. Two methods resting on different functional assumptions, one imposing a parametric curvature,

the other imposing no form at all, rule out the idea of an effect increasing monotonically with disadvantage.

The resulting profile nonetheless deserves to be described precisely, because the two non-significances are not of the same nature. Among the least disadvantaged pupils, the coefficient is near zero and precisely estimated: -0.045 , within an interval $[-0.18; +0.09]$ that rules out any substantial effect. Among the most disadvantaged, the point estimate, -0.244 , is close to the overall estimate, but its interval, $[-0.63; +0.14]$, is too wide to settle anything: it is compatible with the absence of an effect as well as with an effect larger than that of the intermediate tercile. The pairwise comparisons confirm this asymmetry: the rejection of homogeneity rests entirely on the contrast between the low and middle terciles, the high tercile differing significantly from neither of the other two.

One feature of the construction sheds light on this last point. The high tercile gathers all schools with `tipuach` > 15 , a range that encompasses the turning point estimated by the quadratic specification (≈ 23), which, as we saw, is itself very imprecisely located, the confidence interval on $\hat{\delta}_2$ moving it from 14 to 73. This tercile therefore probably mixes schools on either side of the parabola’s vertex, and its coefficient would average the two sides. A finer split in the upper part of the distribution would be needed to isolate the behaviour of the effect at the most extreme levels of disadvantage, but the sample does not permit this with sufficient precision.

Finally, this result holds only in reading: mathematics shows no detectable heterogeneity, whichever method is used.

We therefore conclude that there is non-linear heterogeneity in the effect by level of disadvantage, established in reading and absent in mathematics. For the evaluation of the REP+ class-size halving policy, the implications of this result are asymmetric, and it is important not to over-interpret it in either direction. It invalidates the linear extrapolation underlying the interaction in Table A.9 and the reading of Piketty and Valdenaire [19]: nothing warrants assuming that the effect continues to grow as disadvantage increases, and the gain expected in the most disadvantaged schools cannot therefore be inferred from the slope estimated in intermediate schools. But neither does it establish that the effect would be weak there: it is precisely in that zone, the one targeted by the French policy, that our estimates are least informative.

3 The effect at the bottom of the distribution: the failure rate

3.1 Average effect on the failure rate

The main estimation, like Angrist and Lavy’s work as a whole, bears on the class mean score. Yet a mean can rise without struggling pupils making progress: it suffices for the best pupils to gain a few points. The objective of a class-size reduction policy targeted at disadvantaged backgrounds, such as the REP+ halving, is not primarily to raise the mean, but to reduce the share of struggling pupils. We therefore examine here an outcome absent from the original paper: the effect of class size on the failure rate, measured by the “% failer” variable in the data (`flgrm5` and `flmth5` in fifth grade, `flgrm4` and `flmth4` in fourth grade). This variable gives, at the class level, the share of pupils failing the national test. The threshold is documented by

Lavy (1997) [17], who describes the same testing arrangement: each test included a subset of fifteen questions constituting a basic skills test, and a pupil was counted as failing as soon as they missed three or more. The criterion is therefore demanding, which explains the magnitude of the observed rates. Note that the mean score and the failure rate do not rest on exactly the same items: the latter bears on the fifteen basic questions, whereas the former aggregates some of those questions together with all those from the sixteenth onwards.

We take up the main 2SLS specification, simply substituting this indicator, everything else being unchanged: **classsize** instrumented by $\hat{f}_{sc}$, controls **tipuach** and e_s , standard errors clustered at the school level. The direction of reading is reversed, however, since the indicator is oriented oppositely to the score: a positive coefficient means that one additional pupil *increases* the share of failing pupils.

Table 6: Effect of one additional pupil on the failure rate: OLS and 2SLS

Grade	Subject	Coefficient		coef / σ (failure)		Recall score
		OLS	2SLS	OLS	2SLS	
Fifth	Reading	+0.026	+0.309**	0.20%	2.42%	3.58%
	Maths	-0.029	+0.179	0.20%	1.20%	2.12%
Fourth	Reading	-0.002	+0.064	0.03%	0.70%	1.66%
	Maths	+0.106	+0.226	0.69%	1.46%	0.57%

Notes: dependent variable: share of failing pupils (%). A positive coefficient means that one additional pupil raises this share. Both estimators include the same controls, **tipuach** and e_s ; 2SLS instruments **classsize** with $\hat{f}_{sc}$. The |coef|/ σ columns express the coefficient relative to the standard deviation of the failure rate, so as to make the two estimators comparable. The last column recalls the 2SLS effect on the mean score, in standard deviations. Cluster-robust standard errors at the school level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The results are presented in Table 6. The effect is significant in reading among fifth graders: one additional pupil raises the failure rate by 0.31 points ($p < 0.05$). Relative to the standard deviation of the indicator, this effect (2.42%) is of the same order as the effect on the mean score (3.58%). Reducing class size therefore does not act only on the mean: it also reduces the share of struggling pupils, by a comparable magnitude, which speaks directly to the redistributive objective of the class-size halving policy. As with scores, the effect is concentrated on reading in fifth grade and does not attain significance in mathematics or in fourth grade.

3.2 Testing curvature: the quadratic interaction

We apply to the failure rate the same sequence as for scores. We first test curvature directly, by adding the term **classsize** $\times$ **tipuach**² to the specification. The three endogenous variables are instrumented simultaneously by $\hat{f}_{sc}$, $\hat{f}_{sc} \times \text{tipuach}$ and $\hat{f}_{sc} \times \text{tipuach}^2$; the first stages are strong across all three equations (F between 112 and 214), ruling out any weak-instrument problem.

Table 7: Quadratic interaction `classsize` $\times$ `tipuach`² on the failure rate (fifth grade)

	Reading	Mathematics
<code>classsize</code>	−0.064 (0.147)	−0.034 (0.189)
<code>classsize</code> $\times$ <code>tipuach</code>	+0.0589*** (0.0189)	+0.0319 (0.0195)
<code>classsize</code> $\times$ <code>tipuach</code> ²	−0.00117*** (0.00038)	−0.00060 (0.00041)
First-stage F	214/112/129	214/112/129
Recall, scores: $\hat{\delta}_2 = +0.00051$ *** (read.), $+0.00047$ ** (math.)		

Notes: 2SLS, fifth grade. Dependent variable: failure rate (%). Endogenous: `classsize`, `classsize` $\times$ `tipuach`, `classsize` $\times$ `tipuach`²; instruments: $\hat{f}_{sc}$, $\hat{f}_{sc} \times \text{tipuach}$, $\hat{f}_{sc} \times \text{tipuach}^2$; controls: `tipuach`, `tipuach`², e_s . A negative quadratic term corresponds to a bell-shaped profile (the outcome having the opposite sign to the score). First-stage F : the three values correspond to `classsize`, `classsize` $\times$ `tipuach` and `classsize` $\times$ `tipuach`². Cluster-robust standard errors (school). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The results appear in Table 7. In reading, the quadratic term is significant at 1% ($\hat{\delta}_2 = -0.00117$, $p = 0.002$), with the sign expected for a bell-shaped relationship (negative, since the outcome is inverted relative to the score). This is the striking result: the same curvature as on scores ($\hat{\delta}_2 = +0.00051$ in reading) reappears on a distinct outcome. The effect of class size on the failure rate is maximal around `tipuach` ≈ 25 , very close to the turning point estimated on scores (≈ 23). In mathematics, by contrast, the quadratic term is not significant ($p = 0.137$), just as heterogeneity was already weaker there on scores.

3.3 Tercile decomposition

We apply to the failure rate the same decomposition as to scores: the main 2SLS model is re-estimated separately on each of the three terciles of `tipuach`, and the equality of the three slopes is then subjected to a Wald test. As before, since the terciles are constructed on a school-level variable, no cluster straddles a group and the three estimates are independent: the test is computed directly from their coefficients. The controls retained are those of the main model, `tipuach` and e_s .

Table 8: Effect on the failure rate by tercile of `tipuach` and test of equality (fifth grade)

	Reading			Mathematics		
	Coef. (SE)		95% CI	Coef. (SE)		95% CI
Low ($t \leq 6$)	-0.253 (0.158)		[-0.56 ; +0.06]	-0.029 (0.213)		[-0.45 ; +0.39]
Middle ($6 < t \leq 15$)	+0.739*** (0.271)		[+0.21 ; +1.27]	+0.559 (0.355)		[-0.14 ; +1.26]
High ($t > 15$)	+0.718* (0.402)		[-0.07 ; +1.51]	+0.163 (0.408)		[-0.64 ; +0.96]
<i>Wald test, $H_0 : \alpha_{\text{low}} = \alpha_{\text{mid}} = \alpha_{\text{high}}$</i>						
<i>W statistic</i>	12.87			2.03		
<i>p-value</i>	0.002			0.362		
<i>Conclusion</i>	H_0 rejected			H_0 not rejected		

Notes: 2SLS, instrument $\hat{f}_{sc}$, controls `tipuach` and e_s ; t denotes `tipuach`. Dependent variable: share of failing pupils (%). A *positive* coefficient means that one additional pupil raises the share of failing pupils, i.e. an effect of the same nature as a negative coefficient on the score. Sample sizes: 739, 603 and 630 classes; first-stage F : 121.1, 50.5 and 35.4. Wald test with 2 degrees of freedom, computed from the three separate estimations, whose covariances are null (terciles built on a school-level variable, disjoint clusters). Cluster-robust standard errors at the school level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: Pairwise comparisons between terciles, failure rate (fifth grade)

Comparison	Gap	(SE)	z	Raw p	Bonf. p
<i>Reading</i>					
Low vs middle	-0.992	(0.314)	-3.16	0.002	0.005
Middle vs high	+0.021	(0.485)	+0.04	0.965	1.000
Low vs high	-0.970	(0.432)	-2.25	0.025	0.074
<i>Mathematics</i>					
Low vs middle	-0.589	(0.414)	-1.42	0.155	0.464
Middle vs high	+0.396	(0.541)	+0.73	0.464	1.000
Low vs high	-0.192	(0.460)	-0.42	0.677	1.000

Notes: the gap is computed as the coefficient of the first group named minus that of the second; a negative gap here means the effect is more pronounced in the second group, the failure rate being oriented inversely to the score. The standard error of the gap is obtained by summing variances, the three estimations being independent. The “Bonf. p ” column multiplies the raw p -values by the number of comparisons conducted; values of 1.000 result from the capping. These comparisons are presented to locate the difference; the reference test remains the joint test in Table 8.

The results appear in Table 8. In reading, homogeneity is clearly rejected: $W = 12.87$, 2 df, $p = 0.002$. In mathematics, it is not ($W = 2.03$, $p = 0.362$), in line with what our extension shows for that subject throughout.

A threshold profile rather than a bell shape. The shape of the heterogeneity nonetheless differs from that observed on scores, and Table 9 allows it to be located precisely. On scores, it was the intermediate tercile that stood apart from the other two. Here, the middle and high

terciles are practically identical, $+0.739$ and $+0.718$, their gap being null to all intents and purposes ($p = 0.965$), and it is the low tercile that departs from both, significantly from the middle one ($p = 0.005$ after correction) and marginally from the high one ($p = 0.074$).

In other words, the effect on the failure rate does not describe a bell but a threshold: it is undetectable in the least disadvantaged schools, then substantial and of comparable magnitude in the upper two thirds of the distribution. Note that the coefficient of the low tercile has the opposite sign to the expected effect; it is not distinguishable from zero, its confidence interval amply covering that value, and this sign should not be interpreted.

This is the most important result of our extension, because it bears precisely where the estimation on scores remained silent. In the most disadvantaged tercile, the coefficient on the mean score was -0.244 with an interval too wide to settle anything. On the failure rate, the same population yields an effect of $+0.718$, significant at the 10% level, i.e. about 5.6% of a standard deviation of the indicator per pupil, a magnitude that is far from negligible.

The divergence between the two measures is not contradictory: it is instructive. Mean score and failure rate do not, moreover, rest on exactly the same items (cf. § 3.1), and above all they do not measure the same thing. A class mean can remain stable while the composition of the distribution shifts. What these estimates suggest is that, in very disadvantaged schools, reducing class size acts on the share of struggling pupils without that effect being detectable on the mean. Yet this is exactly the objective pursued by a class-size halving policy targeted at priority education.

Robustness and reservations. The result does not depend on the choice of controls: estimated with the e_s control alone, the test still rejects homogeneity in reading, with a higher p -value ($p = 0.016$). We retain the specification including `tipuach`, disadvantage being a plausible determinant of the failure rate whose omission would expose us to omitted variable bias.

Three reservations nonetheless accompany this result. First, it holds only in reading; mathematics shows no detectable heterogeneity, which is consistent with the rest of our extension but limits the scope of the conclusion. Second, the significance of the high tercile is attained only at the 10% level, and its confidence interval remains wide: the magnitude of the effect there is poorly pinned down, even if its existence is better established than on scores. Third, our extension chains together a fair number of tests, which mechanically increases the risk that at least one of them comes out significant by chance, a reservation all the more pertinent in that the profile obtained here does not exactly reproduce that of the scores.

We therefore conclude that reducing class size lowers the share of pupils failing in reading, and that this effect is concentrated in the two most disadvantaged thirds of the distribution, without it being possible to distinguish these two groups from one another.

4 Characterising the compliers

The instrumental variables estimator does not identify an average effect across all schools, but a local effect (LATE), driven solely by schools whose class size actually responds to Maimonides' rule, namely those located near the thresholds of 40, 80 and 120. The scope of the

result therefore depends on whether these pivotal schools constitute a particular subset or are representative of the population studied. To assess this, we compare the average characteristics of schools near the thresholds (± 5 windows, the *compliers*) with those of schools far from them. This analysis appears neither in Angrist and Lavy (1999) nor in Angrist et al. (2019).

Table 10: Characteristics of schools near the thresholds (compliers) vs. far from them (fifth grade)

Variable	Near (compliers)	Far	Gap	<i>p</i>
% disadvantaged (<i>tipuach</i>)	13.59	14.25	−0.66	0.348
Cohort enrolment	76.38	78.15	−1.77	0.303
Class size	30.82	29.67	+1.15	0.002
Reading score	74.49	74.35	+0.15	0.731
Mathematics score	67.03	67.36	−0.33	0.530
Failure rate (reading)	24.26	24.91	−0.65	0.354
Number of fifth-grade classes	2.66	2.67	−0.01	0.805
<i>N</i>	471	1,547		

Notes: “near the thresholds” = schools whose cohort enrolment falls within a ± 5 window around 40, 80 or 120. Gap = mean (near) − mean (far). *p*: *t*-test of the difference in means (unequal variances). Only class size differs significantly, for a mechanical reason related to position relative to the threshold.

Schools near the thresholds, which represent 23% of the sample, do not differ significantly from the others on *any* of the socio-educational dimensions considered (Table 10): share of disadvantaged pupils, scores, failure rate, cohort size and number of classes all display non-significant gaps. The only significant difference is mechanical: *compliers* have a slightly larger class size (+1.15 pupils, $p = 0.002$), which is to be expected since, near the thresholds and before they are crossed, classes are close to their maximum. On observable characteristics, the estimated LATE is therefore representative of the sample’s school population: the schools driving identification are not atypical.

Part V. Revisiting 1999 in the light of 2019

1 What the replication and the extension leave open

The replication in Part III recovers with precision the coefficients of the *first stage*, the reduced form and 2SLS, and the extension in Part IV has taken that examination further on the terrain of internal validity: the pivotal schools driving identification differ from the others on no observable socio-educational characteristic (Table 10), and heterogeneity by level of disadvantage, once formally tested, proves non-linear in reading and undetectable in mathematics. These checks all bear on a single sample, that of Israel in 1991. None tells us whether the coefficient obtained holds outside that point in time. That is precisely the question opened up by the authors’ return to their own data.

2 A robust zero

Angrist et al. (2019) [4] reapply the same strategy to 240,310 fifth-grade pupils tested between 2002 and 2011 under the GEMS programme. The 2SLS estimates lie between -0.03 and $+0.03$, with standard errors on the order of 0.03 to 0.04: precisely estimated zeros, against the -0.275 and -0.229 we replicated. The coefficient on `tipuach` (SES) remains, for its part, very clear, which rules out a lack of statistical power.

The main threat is enrolment manipulation: the histogram of enrolments declared in November displays spikes just above 40 and 80, principals having an interest in triggering the opening of an additional class. Two checks rule this out: an alternative instrument based on enrolments imputed from dates of birth produces the same zeros, and the correlation between instrument and SES, real but modest, disappears as soon as two school-level covariates are introduced. In the 1991 data, manipulation existed but remains econometrically innocuous: the so-called *donut* strategy, which removes schools with enrolment between 39 and 41 pupils, yields -0.234 against -0.277 on the full sample. This test converges with our characterisation of the *compliers*: the schools the *donut* removes are, up to the window, those we showed differ from the others on no observable. Removing them does not change the estimate, precisely because they are not atypical.

3 What 2019 changes, read from our extension

The rereading proposed by the authors weakens the reach of 1991 without contesting its mechanics. They recall that the effects were already markedly weaker in fourth grade and absent from the 1992 sample, and that moving from the Moulton correction to school-level clustering, the very substitution made in our replication (§ 3.2), increases the estimated variance of the original coefficients. Our own standard errors, slightly larger than the paper's, are the direct trace of this point.

No explanation for the disappearance is conclusive taken in isolation: median class size falls from 31 to 28 pupils, but with no discernible annual trend; growing test preparation should leave a temporal signature that is likewise not observed. That leaves the hypothesis of supplementary resources, namely additional teachers funded by parents in high socio-economic status schools. It is worth setting alongside the profile obtained in Part IV: it is in the least disadvantaged tercile that we found a precisely estimated zero (-0.045), and in the upper two thirds that the effect on the failure rate becomes substantial. Both results point to the same mechanism, class size ceasing to act where other resources take over, and suggest diminishing returns.

One reservation, finally, that our extension allows us to formulate and that the paper does not address. Like those of 1999, the 2019 estimates bear on the class mean score. Yet we have shown that a mean can remain silent where the failure rate moves: that is exactly what happens in our most disadvantaged tercile, where the score gave -0.244 within an unusable interval while the failure rate gave $+0.718$. The zero for 2002–2011 is a zero on the mean; it says nothing about what happens at the bottom of the distribution, which is precisely the target of the REP+ policy.

Conclusion

The replication reproduces with precision Angrist and Lavy’s coefficients, -0.275 in reading and -0.229 in mathematics for fifth graders. It validates the implementation of the procedure and the consistency of the data, without thereby demonstrating the exclusion restriction, which remains an argued and untestable assumption.

The extension establishes three things. First, the linearity assumption does not survive scrutiny: the quadratic specification rejects it at 1%, and the Wald test on the three terciles rejects homogeneity in reading ($W = 8.85$, $p = 0.012$), the contrast opposing the least disadvantaged schools, where the effect is a precisely estimated zero, to those of intermediate disadvantage, where it reaches -0.512 . Second, a limitation: on scores, the most disadvantaged tercile, the one that bears directly on French policy, gives -0.244 within an interval compatible with the absence of an effect as well as with an effect larger than the intermediate tercile’s. The most relevant zone is also the one where estimates are least informative. Third, a result that lifts this limitation by changing the measure: on the failure rate, the same population gives $+0.718$, significant, and homogeneity is rejected there more clearly still ($p = 0.002$). The effect then describes not a bell but a threshold, undetectable in the least disadvantaged schools and then substantial in the upper two thirds.

This last point governs how the French evaluations should be read. Andreu et al. (2021) observe exactly the same phenomenon: in mathematics, the share of REP+ pupils below the control group’s first quintile falls from 21.4% to 15.9%, i.e., by their own conversion, about 14% of a standard deviation against the 8% descriptive effect on mean scores. They write that the improvement appears stronger for struggling pupils, without however establishing it: the quintile analysis remains descriptive and is the subject of no difference-in-differences estimation. Our extension supplies what it lacked, a test, and recovers the finding in an entirely different context and identification strategy. El Bah (2026), evaluating the same cohorts at the start of lower secondary school, does examine one bottom-of-distribution indicator, but on a dimension where the authors themselves note a clear violation of the parallel trends assumption.

This thesis does not in itself allow one to conclude whether the REP+ class-size halving will be effective; that question goes beyond the scope of a replication based on Israeli data from 1991. It does, however, shift the question. The argument most often invoked in support of this policy, that of an effect increasing with disadvantage, does not hold in the linear form ascribed to it. But the failure of this extrapolation does not condemn the policy; it condemns the way its benefits are expected. Where the mean score says nothing more in the most disadvantaged tercile, the failure rate reveals a substantial effect there, and it is precisely this objective that a policy targeted at priority education pursues.

What remains is to measure what the available evaluations measure poorly: the bottom of the distribution, with the same rigour as the mean, the two French attempts each failing on one side, one remaining descriptive, the other resting on an invalid counterfactual; and the effect of class size isolated from the accompanying measures, which Andreu et al. (2021) acknowledge inflate their estimate. This is a modest conclusion, but it bears on what the data actually show, in a field where even the literature’s most solidly established study has not withstood the test of time in its own country of origin.

References

- [1] Andreu, E., Douat, É., Fabre, M., & Fenet, S. (2021). Évaluation du dédoublement des classes de CP et de CE1 dans les réseaux d'éducation prioritaire renforcée. *Note d'information*, no. 21.19, DEPP.
- [2] Angrist, J. D., & Lavy, V. (1999). Using Maimonides' rule to estimate the effect of class size on scholastic achievement. *The Quarterly Journal of Economics*, 114(2), 533–575.
- [3] Angrist, J. D., Battistin, E., & Vuri, D. (2017). In a small moment: Class size and moral hazard in the Italian Mezzogiorno. *American Economic Journal: Applied Economics*, 9(4), 216–249.
- [4] Angrist, J. D., Lavy, V., Leder-Luis, J., & Shany, A. (2019). Maimonides' rule redux. *American Economic Review: Insights*, 1(3), 309–324.
- [5] Bouguen, A., Grenet, J., & Gurgand, M. (2017). Does class size influence student progress at school? *IPP Policy Brief*, no. 28.
- [6] Bressoux, P., Kramarz, F., & Prost, C. (2009). Teachers' training, class size and students' outcomes: Learning from administrative forecasting mistakes. *The Economic Journal*, 119(536), 540–561.
- [7] Bressoux, P., & Lima, L. (2011). La place de l'évaluation dans les politiques éducatives : le cas de la taille des classes à l'école primaire en France. In G. Felouzis & S. Hanhart (eds.), *Gouverner l'éducation par les nombres ?* (pp. 99–123). Brussels: De Boeck.
- [8] Chetty, R., Friedman, J. N., Hilger, N., Saez, E., Schanzenbach, D. W., & Yagan, D. (2011). How does your kindergarten classroom affect your earnings? Evidence from Project STAR. *The Quarterly Journal of Economics*, 126(4), 1593–1660.
- [9] El Bah, A. (2026). Évaluation du dédoublement des classes de CP et CE1 en REP+ à l'entrée en sixième. Working paper, DEPP.
- [10] Finn, J. D., & Achilles, C. M. (1990). Answers and questions about class size: A statewide experiment. *American Educational Research Journal*, 27(3), 557–577.
- [11] Fredriksson, P., Öckert, B., & Oosterbeek, H. (2013). Long-term effects of class size. *The Quarterly Journal of Economics*, 128(1), 249–285.
- [12] Gelman, A., & Stern, H. (2006). The difference between “significant” and “not significant” is not itself statistically significant. *The American Statistician*, 60(4), 328–331.
- [13] Hoxby, C. M. (2000). The effects of class size on student achievement: New evidence from population variation. *The Quarterly Journal of Economics*, 115(4), 1239–1285.
- [14] Jepsen, C., & Rivkin, S. (2009). Class size reduction and student achievement: The potential tradeoff between teacher quality and class size. *Journal of Human Resources*, 44(1), 223–250.
- [15] Kraft, M. A. (2020). Interpreting effect sizes of education interventions. *Educational Researcher*, 49(4), 241–253.
- [16] Krueger, A. B. (1999). Experimental estimates of education production functions. *The Quarterly Journal of Economics*, 114(2), 497–532.
- [17] Lavy, V. (1997). *Hevdelim be-Masabim u-ve-Hasagim be-Hinukh ha-'Aravi be-Yisrael* [Differences in resources and outcomes in Arab education in Israel, in Hebrew]. Jerusalem: Floersheimer Institute for Policy Studies.
- [18] Moulton, B. R. (1986). Random group effects and the precision of regression estimates. *Journal of Econometrics*, 32(3), 385–397.
- [19] Piketty, T., & Valdenaire, M. (2006). L'impact de la taille des classes sur la réussite scolaire dans les écoles, collèges et lycées français. *Les Dossiers*, no. 173, DEP.
- [20] Shen, T., & Konstantopoulos, S. (2021). Class size effects on reading and mathematics achievement in Europe: Evidence from TIMSS. *Educational Research and Evaluation*, 27(1–2), 1–31.
- [21] Urquiola, M. (2006). Identifying class size effects in developing countries: Evidence from rural Bolivia. *The Review of Economics and Statistics*, 88(1), 171–177.

Appendices

Table A.1: Comparative synthesis of the studies reviewed

Impact of one fewer student on academic performance — updated IPP literature review				
Study	Country	Impact per student removed (% of a standard deviation of the score)	Subject / Grade	Empirical method
A. Primary school				
Krueger (1999)	U.S.A	2,66	Reading and maths (kindergarten)	Randomised experiment
Krueger (1999)	U.S.A	3,31	Reading and maths (grade 1)	Randomised experiment
Krueger (1999)	U.S.A	1,85	Reading and maths (grade 2)	Randomised experiment
Krueger (1999)	U.S.A	2,44	Reading and maths (grade 3)	Randomised experiment
Hoxby (2000)	U.S.A	0,00	Maths (grade 3)	Quasi-experiment
Hoxby (2000)	U.S.A	0,01	Reading (grade 3)	Quasi-experiment
Hoxby (2000)	U.S.A	0,00	Writing (grade 3)	Quasi-experiment
Hoxby (2000)	U.S.A	0,00	Maths (grade 5)	Quasi-experiment
Hoxby (2000)	U.S.A	0,00	Reading (grade 5)	Quasi-experiment
Hoxby (2000)	U.S.A	0,01	Writing (grade 5)	Quasi-experiment
Angrist and Lavy (1999)	Israel	0,57	Maths (grade 4)	Class-opening thresholds
Angrist and Lavy (1999)	Israel	1,66	Reading (grade 4)	Class-opening thresholds
Angrist and Lavy (1999)	Israel	2,39	Maths (grade 5)	Class-opening thresholds
Angrist and Lavy (1999)	Israel	3,59	Reading (grade 5)	Class-opening thresholds
Piketty and Valdenaire (2006)	France	2,79	Maths (grade 2)	Class-opening thresholds
Piketty and Valdenaire (2006)	France	2,19	French (grade 2)	Class-opening thresholds
Bressoux et al. (2009)	France	2,70	Maths (grade 3)	Quasi-experiment
Bressoux et al. (2009)	France	2,60	Reading (grade 3)	Quasi-experiment
Bressoux and Lima (2011)	France	2,20	Maths and French (grade 1)	Non-randomised experiment
Fredriksson et al. (2013)	Sweden	3,30	Logical and verbal skills (grades 4–6)	Class-opening thresholds
Iversen and Bonesrønning (2013)	Norway	0,46	Maths and reading (grade 4)	Class-opening thresholds
Andreu et al. / DEPP (2021) ★	France	1,30	French (grade 1)	Difference-in-differences
Andreu et al. / DEPP (2021) ★	France	2,00	Maths (grade 1)	Difference-in-differences
Andreu et al. / DEPP (2021) ★	France	N.S.	French (grade 2)	Difference-in-differences
Andreu et al. / DEPP (2021) ★	France	N.S.	Maths (grade 2)	Difference-in-differences
Angrist et al. (2019) ★	Israel	N.S.	Maths (grade 5)	Class-opening thresholds
Angrist et al. (2019) ★	Israel	N.S.	Hebrew (grade 5)	Class-opening thresholds
Ei Bah / DEPP (2026) ★	France	N.S.	French (entry to middle school)	Difference-in-differences
Ei Bah / DEPP (2026) ★	France	N.S.	Maths (entry to middle school)	Difference-in-differences
Jepsen and Rivkin (2009) ★	United States	0.56 to 0.96	Maths (grades 2–4)	Difference-in-differences
Jepsen and Rivkin (2009) ★	United States	0.35 to 0.63	Reading (grades 2–4)	Difference-in-differences
Urquiola (2006) ★	Bolivia	3,25	Spanish (grade 2, rural schools)	Class-opening thresholds
Urquiola (2006) ★	Bolivia	2,38	Maths (grade 2, rural schools)	Class-opening thresholds
B. Lower secondary school				
Piketty and Valdenaire (2006)	France	0,80	Maths (grade 9)	Class-opening thresholds
Piketty and Valdenaire (2006)	France	1,14	French (grade 9)	Class-opening thresholds
Piketty and Valdenaire (2006)	France	1,63	History and geography (grade 9)	Class-opening thresholds
Leuven et al. (2008)	Norway	0,01	Maths (grade 7)	Class-opening thresholds
Leuven et al. (2008)	Norway	0,01	Norwegian (grade 7)	Class-opening thresholds
Shen & Konstantopoulos (2021) ★	Romania	5 to 9	Maths, physics, chemistry, earth science (grade 8) — 2003 only Biology not sig.; 2007 and 2011 not sig.	Class-opening thresholds
Shen & Konstantopoulos (2021) ★	Lithuania	N.S.	All subjects, all years (2003, 2007, 2011)	Class-opening thresholds
Shen & Konstantopoulos (2021) ★	Slovenia	N.S.	All subjects, all years (2003, 2007, 2011)	Class-opening thresholds
Shen & Konstantopoulos (2021) ★	Hungary	N.I.	Invalid instrument ($F < 10$) — results not interpretable	Class-opening thresholds

★ : new entry

Note on Angrist and Lavy (1999). Unlike Bouguen et al. (2017), whose table draws on Table VI of the original paper (discontinuity sample, dummy instruments) normalised by an implicit standard deviation of 10, the figures reported here follow the convention of the present table throughout: the 2SLS coefficient of the main specification (Table IV, linear enrolment control) divided by the standard deviation of class mean scores (Tables A.3 and A.4).

A. Descriptive statistics

Table A.2: Descriptive statistics for the raw fifth-grade data

	schlcode	classsize	c_size	avgverb	avgmath	tipuach	c_leom	c_pik
count	2029.00	2029.00	2029.00	2024.00	2024.00	2029.00	2029.00	2029.00
mean	39606.33	29.95	77.87	74.45	67.32	14.09	1.00	1.25
std	15282.75	6.60	39.06	8.08	10.03	13.48	0.00	0.43
min	11005.00	5.00	5.00	34.80	0.00	0.00	1.00	1.00
25%	31037.00	26.00	50.00	69.86	61.14	4.00	1.00	1.00
50%	41212.00	31.00	72.00	75.43	67.80	10.00	1.00	1.00
75%	51203.00	35.00	100.00	79.85	74.09	19.00	1.00	1.00
max	61365.00	47.00	226.00	187.61	181.25	76.00	1.00	2.00

Table A.3: Descriptive statistics for fifth-grade classes after processing

	classsize	c_size	func1	avgverb	avgmath	tipuach
count	2018.00	2018.00	2018.00	2018.00	2017.00	2018.00
mean	29.94	77.74	30.96	74.38	67.29	14.1
std	6.55	38.82	6.11	7.68	9.60	13.5
min	8.00	8.00	8.00	34.80	27.69	0.0
25%	26.00	50.00	27.00	69.85	61.10	4.0
50%	31.00	72.00	31.67	75.39	67.78	10.0
75%	35.00	100.00	35.67	79.84	74.08	19.0
max	44.00	226.00	40.00	93.86	93.93	76.0

Table A.4: Descriptive statistics for fourth-grade classes after processing

	classsize	c_size	func1	avgverb	avgmath	tipuach
count	2049.00	2049.00	2049.00	2049.00	2049.00	2049.00
mean	30.28	78.30	31.09	72.49	68.86	13.83
std	6.33	37.71	5.87	7.99	8.77	13.35
min	7.00	8.00	8.00	24.29	21.41	0.00
25%	26.00	51.00	27.50	67.67	63.59	4.00
50%	31.00	74.00	32.00	73.33	69.33	9.00
75%	35.00	101.00	35.50	78.21	74.97	19.00
max	44.00	225.00	40.00	94.90	94.09	76.00

B. Replication results

Table A.5: OLS results

A. Reading			
	(1)	(2)	(3)
	<i>classize only</i>	<i>+ tipuach</i>	<i>+ tipuach + enrolment</i>
classize	0.2213***	−0.0315	−0.0246
Standard error	(0.0335)	(0.0262)	(0.0329)
<i>t</i> -statistic	6.60	−1.20	−0.75
<i>p</i> -value	0.000	0.229	0.455
B. Mathematics			
	(4)	(5)	(6)
	<i>classize only</i>	<i>+ tipuach</i>	<i>+ tipuach + enrolment</i>
classize	0.3220***	0.0757**	0.0192
Standard error	(0.0402)	(0.0358)	(0.0421)
<i>t</i> -statistic	8.02	2.12	0.45
<i>p</i> -value	0.000	0.034	0.649

Notes: reported coefficients are the estimated effect of **classize** on mean reading and mathematics scores. **tipuach** denotes the school’s share of disadvantaged students, labelled SES in Angrist and Lavy; “enrolment” denotes cohort size e_s . Standard errors, corrected for within-school correlation, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.6: First stage and reduced forms

	First stage		Reduced form		Reduced form	
	classize		avgverb		avgmath	
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>without e_s</i>	<i>with e_s</i>	<i>without e_s</i>	<i>with e_s</i>	<i>without e_s</i>	<i>with e_s</i>
func1	0.7035***	0.5415***	−0.1114***	−0.1490***	−0.0090	−0.1240**
Standard error	(0.0253)	(0.0367)	(0.0286)	(0.0389)	(0.0405)	(0.0515)
<i>t</i> -statistic	27.83	14.76	−3.89	−3.83	−0.22	−2.41
<i>p</i> -value	0.000	0.000	0.000	0.000	0.823	0.016

Notes: columns (1) and (2) report the first stage, that is, the regression of **classize** on **func1**. Columns (3) to (6) report the reduced forms, that is, the direct regressions of scores on **func1**. All columns include the socio-economic control (**tipuach**). Columns labelled “with e_s ” add a linear control for cohort enrolment. Standard errors, corrected for within-school correlation, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.7: 2SLS results

Panel A – Fifth grade					
	(1) OLS	(2) 2SLS without e_s	(3) 2SLS with e_s	(4) 2SLS with $\log(e_s)$	(5) 2SLS with e_s^2
<i>Controls</i>					
tipuach (SES)	Yes	Yes	Yes	Yes	Yes
Functional form of e_s	—	none	linear	log	quadratic
<i>Reading</i>					
classsize	−0.0315	−0.1583***	−0.2751***	−0.3040***	−0.2588***
Standard error	(0.0262)	(0.0416)	(0.0758)	(0.1012)	(0.0937)
<i>t</i> -statistic	−1.20	−3.80	−3.63	−3.01	−2.76
<i>p</i> -value	0.229	0.000	0.000	0.003	0.006
<i>Mathematics</i>					
classsize	0.0757**	−0.0129	−0.2291**	−0.3084**	−0.2600**
Standard error	(0.0358)	(0.0577)	(0.0984)	(0.1317)	(0.1231)
<i>t</i> -statistic	2.12	−0.22	−2.33	−2.34	−2.11
<i>p</i> -value	0.034	0.823	0.020	0.019	0.035
Panel B – Fourth grade					
	(1) OLS	(2) 2SLS without e_s	(3) 2SLS with e_s	(4) 2SLS with $\log(e_s)$	(5) 2SLS with e_s^2
<i>Controls</i>					
tipuach (SES)	Yes	Yes	Yes	Yes	Yes
Functional form of e_s	—	none	linear	log	quadratic
<i>Reading</i>					
classsize	−0.0530*	−0.1100***	−0.1329**	−0.0973	−0.0739
Standard error	(0.0276)	(0.0404)	(0.0608)	(0.0720)	(0.0683)
<i>t</i> -statistic	−1.92	−2.72	−2.18	−1.35	−1.08
<i>p</i> -value	0.055	0.006	0.029	0.177	0.279
<i>Mathematics</i>					
classsize	0.0548	0.0487	−0.0497	−0.0590	−0.0328
Standard error	(0.0355)	(0.0520)	(0.0747)	(0.0889)	(0.0845)
<i>t</i> -statistic	1.54	0.94	−0.67	−0.66	−0.39
<i>p</i> -value	0.123	0.349	0.506	0.507	0.698

Notes: all columns include the school-level socio-economic control (**tipuach**). Columns (2) to (5) differ only in the functional form of the control for cohort enrolment e_s . The 2SLS columns use Maimonides' rule as instrument. Columns (2), (3) and (5) correspond respectively to columns (1), (2) and (3) of Table IV in Angrist and Lavy (1999); column (4) is an addition. Column (3) is the main specification. Standard errors, corrected for within-school correlation, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.8: 2SLS results – discontinuity sample ± 5 , binary instruments

	Fifth grade ($N = 471$)		Fourth grade ($N = 415$)	
	Reading	Maths	Reading	Maths
<i>A. Without SES control (seg₁ + seg₂ only)</i>				
classsize	−0.654**	−0.568*	−0.358**	−0.134
Standard error	(0.329)	(0.342)	(0.176)	(0.184)
<i>p</i> -value	0.047	0.097	0.042	0.467
<i>B. With SES control (tipuach + seg₁ + seg₂)</i>				
classsize	−0.687***	−0.596**	−0.175	+0.018
Standard error	(0.264)	(0.291)	(0.137)	(0.151)
<i>p</i> -value	0.009	0.041	0.201	0.905

Notes: binary instruments $z_1 = \mathbf{1}[41 \leq e_s \leq 45]$, $z_2 = \mathbf{1}[81 \leq e_s \leq 85]$, $z_3 = \mathbf{1}[121 \leq e_s \leq 125]$. Controls: window dummies seg_1 , seg_2 (panel A); plus **tipuach** (panel B). Cluster-robust standard errors at the school level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.9: 2SLS results – `classsize` × `tipuach` interaction

	Fifth grade		Fourth grade		Pooled	
	Reading	Maths	Reading	Maths	Reading	Maths
<code>classsize</code>	−0.155**	−0.079	−0.101	+0.019	−0.120**	−0.018
Standard error	(0.072)	(0.104)	(0.065)	(0.082)	(0.053)	(0.070)
<i>p</i> -value	0.031	0.445	0.124	0.817	0.025	0.793
<code>classsize</code> × <code>tipuach</code>	−0.008**	−0.010**	−0.002	−0.005	−0.005*	−0.007**
Standard error	(0.004)	(0.005)	(0.003)	(0.004)	(0.003)	(0.004)
<i>p</i> -value	0.037	0.026	0.502	0.221	0.073	0.038

Notes: endogenous variables: `classsize`, `classsize`×`tipuach`. Instruments: $\hat{f}_{sc}$, $\hat{f}_{sc}$ × `tipuach`. Controls: `tipuach`, e_s (linear). Pooled: plus a `grade4` dummy (= 1 for fourth grade). Cluster-robust standard errors at the school level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C. Implementation

The two utility functions that serve as the building block for all the estimates are `ols_cluster` for OLS regressions and `iv_cluster` for 2SLS estimates.

Concretely, `ols_cluster` runs an OLS regression via `statsmodels` with `cov_type='cluster'` and `schlcode` as the clustering variable: each school is treated as a group whose residuals may be arbitrarily correlated. The function `iv_cluster` does the same for 2SLS via `linearmodels`, with `cov_type='clustered'`.

In practice, the two approaches, analytical Moulton and *cluster-robust*, converge to very similar results: the coefficients are identical across all specifications, and the standard errors differ only by a few basis points, with the *cluster-robust* estimator being slightly more conservative. This gap is expected and does not call into question the validity of the replication.

D. Code and data

The full replication code and the data files are available at:

[https://1drv.ms/u/c/b01b2a7a4a43ed55/
IQCaiqPv-ycQR6iY3nNXDvXTAXmnKbzw6FN7FeeJoJu4s8s](https://1drv.ms/u/c/b01b2a7a4a43ed55/IQCaiqPv-ycQR6iY3nNXDvXTAXmnKbzw6FN7FeeJoJu4s8s)